QUALIFICATION STANDARD
AND PLAN (QSP)

Basic Occupation Qualification
NQual (AIRY)

MITE ID
Sultan Phase: 109394
Applications Phase: 123626
Stability Phase: 123628
EODC Phase: 123627
Afloat Phase: 109395
Qualification Board: 109396

Regular Force: MS ENG 00345
Training Authority (TA): Commander Naval Personnel and Training Group (CNPTG)
Curriculum Control Authority (CCA): HMCS Venture

MARPAC RDIMS # 1342097
Chap 1 & 2 approved 17 09 2004
Chap 3 & 4 approved 27 07 2023

RECORD OF AMENDMENTS

1. This QSP is developed under the authority of the Royal Canadian Navy (RCN) through the Training Authority (TA), Naval Personnel and Training Group (NPTG) and the Commanding Officer of the HMCS VENTURE. The approval or revision date should appear in the footer of every page as an additional means of verification. Details on the change(s) must be amplified in Annex J "Course History".

Revision	Change Description	Date	Authority
1.0	Initial Version	17 SEP 04	DMTE
1.1			CNPTG Chap. 1 & 2
			CO NTDC Chap. 3 & 4
1.2	Changes to OJPR	2017	CO NTDC(P) Chap. 3 & 4
2.0	New QSP format, new OJPR.	27 Jul 2023	CO HMCS Venture

NOTE:

Personnel programming courses must ensure that their hard copy of this document is current and identical to the electronic copy on the QSP SharePoint Site. See above for current changes.

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FOREWORD

1. This training document is developed under the authority of the Commander Royal Canadian Navy and in accordance with the following references:

a. A-P9-050 Series, Canadian Forces Manual of Individual Training and Education (CFITES)

b. A-PD-055-002/PP-003 Canadian Forces Officer General Specification 20 Nov 2020

c. A-PD-055-002/PP-001 MS ENG MOSID 00345 Job Based Occupation Specification for Marine Systems Engineering Officer 22 Aug 2019, and

d. NAVORD 4500-0 Structured Delegation, and NPTORD 4500-22 Qualification Standard and Plan Production and Management.

2. This QSP consists of specific Performance Objectives (POs) in Chapter 2 and Lesson Specifications and Enabling Objectives (EOs) in Chapter 4. A trainee will achieve the qualification when the standards listed in the POs have been met.

3. This QSP publication is effective on receipt and supersedes MS ENG BOQ QSP revision 1.2, dated 2017. This QSP is designed to assist in developing, conducting, evaluating, and validating the Individual Training and Education (IT&E) required for training.

4. The QS at Chapters 1 and 2 were developed by the Qualification Standard Writing Board (QSWB) convened in 25 February to 22 March 2002 at CFNES Halifax working in collaboration with subject matter experts drawn from participating organizations. The subsequent TP at Chapters 3 and 4 were developed by the Training Plan Writing Board (TPWB) convened 27 January to 21 February 2003 at CFNES Halifax working in collaboration with subject matter experts drawn from participating organizations. All aspects of this document were prepared in accordance with the concept of training outlined in A-P9-050 Series, Canadian Forces Manual of Individual Training and Education. Full details of the WBs are contained within Annex D and L.

5. The criteria for major and minor amendments, and the approval process, are identified in NPTORD 4500-22.

6. In circumstances where the QS and/or TP cannot be adhered to, submission of a Training Waiver in accordance with NPTORD 4500-22 is required.

QUALIFICATION STANDARD

CHAPTER 1 – QUALIFICATION STANDARD

AIM

1. The aim of this QSP is to establish the requirements and training for the Marine System Engineering Officer Basic Occupation Qualification to perform their duties as a Marine System Engineering Officer. Possession of this qualification will allow personnel to (provide a brief description of what they will be able to do).

GRANTING OF NQUAL

2. A trainee will achieve the qualification (MITEs: 109394, 109395, 109396, NQUAL AIRY) when the standards listed in the Performance Objectives (POs) have been met.

CO-QUALIFICATION

3. In addition to the performance objectives specific to this training, the following additional qualifications are included in this training program to meet the requirements of the job. The MITE and NQuals associated with these qualifications shall be granted upon completion of the training program and shall be listed within the course report and will be delivered IAW the following:

a. Naval Engineering Common (Ashore 119913 / Afloat 119912);

b. Officer of the Day qualification – Ashore (108364);

c. Officer of the Day qualification – Afloat (109030);

d. Defence Resource Management Information System (DRMIS) (119096, 119097, 119098);

e. Maritime Warfare Basic (108163);

f. A Qualification Board (PC 001); and

g. Designated TEs incorporating material from both NEC and MS Eng OJPRs (109387)

PREREQUISITES

4. To be selected for this course the loading authority must ensure that trainees meet the following prerequisites:

a. Possess an acceptable Baccalaureate Degree in Science or Engineering IAW CFAO 6-1 if they are a Regular Officer Training Plan (ROTP), Direct Entry Officer (DEO), University Training Plan for Non-Commissioned

Members (UTPNCM), or Special Commissioning Plan (SCP) entry officer;

b. Have completed or have been waived the applicable Academic Bridging course if they are a Commissioned From the Ranks Program (CFRP) entry officers; and

c. Have completed or have been waived the Naval Warfare Officer II course (119728); and

d. Have completed or have been waived the Naval Engineering Indoctrination (NEI) (109286).

CHAPTER 2 – PERFORMANCE OBJECTIVES

PO 001 - MANAGE THE OPERATION OF MAIN PROPULSION SYSTEMS

1. Performance: Manage the Operation of Main Propulsion Systems

2. Conditions:

   a. Given:

      (1) Operational Ship;

      (2) References;

      (3) Engineering Staff Support; and

      (4) Integrated Logistics Support (ILS);.

   b. Denied: N/A.

   c. Environment:

      (1) All Conditions, day or night.

3. Standard: The MS ENG shall manage the operation of main propulsion systems using a total ship's systems integration approach:

   a. IAW:

      (1) CFTO's;

      (2) NEM;

      (3) SSOs; and

      (4) Relevant engineering directives.

4. Reference: TBD.

5. Task Numbers: OT0135, OT0136, DT0070, DT0065, GT0055, GT0060, DT0050, GT0075 and GT0090.

6. Remarks: nil.

PO 002 - MANAGE THE OPERATION OF AUXILIARY SYSTEMS

1. Performance: Manage the Operation of Auxiliary Systems

2. Conditions:

   a. Given:

      (1) Operational Ship;

      (2) References;

      (3) Engineering Staff Support; and

      (4) Integrated Logistics Support (ILS);

   b. Denied: N/A

   c. Environment:

      (1) All conditions, day or night.

3. Standard: The MS ENG shall manage the operation of auxiliary systems using a total ship's systems integration approach.

   a. IAW:

      (1) CFTO's;

      (2) NEM;

      (3) SSOs; and

      (4) Relevant engineering directives.

4. Reference:

   a. TBD.

5. Task Numbers: OT0140, OT0141, DT0070, DT0065, GT0055, GT0060, DT0050, GT0075 and GT0090.

6. Remarks: nil.

PO 003 - MANAGE THE OPERATION OF CONTROL SYSTEMS

1. Performance: Manage the Operation of Control Systems

2. Conditions:

   a. Given:

      (1) Operational Ship;

      (2) References;

      (3) Engineering Staff Support; and

      (4) Integrated Logistics Support (ILS);.

   b. Denied: N/A

   c. Environment:

      (1) All Conditions, day or night.

3. Standard: The MS ENG shall manage the operation of control systems using a total ship's systems integration approach:

   a. IAW:

      (1) CFTO's;

      (2) NEM;

      (3) SSOs; and

      (4) Relevant engineering directives.

4. Reference:

   a. TBD.

5. Task Numbers: OT0145, OT0146, DT0070, DT0065, GT0055, GT0060, DT0050, GT0075 and GT0090.

6. Remarks: nil.

PO 004 - MANAGE THE OPERATION OF ELECTRICAL SYSTEMS

1. Performance: Manage the Operation of Electrical Systems

2. Conditions:

   a. Given:

      (1) Operational Ship;

      (2) References;

      (3) Engineering Staff Support; and

      (4) Integrated Logistics Support (ILS);.

   b. Denied: N/A

   c. Environment:

      (1) All Conditions, day or night.

3. Standard: The MS ENG shall manage the operation of electrical systems using a total ship's systems integration approach:

   a. IAW:

      (1) CFTO's;

      (2) NEM;

      (3) SSOs; and

      (4) Relevant engineering directives.

4. Reference:

   a. TBD

5. Task Numbers: OT0135, OT0136, DT0070, DT0065, GT0055, GT0060, DT0050, GT0075 and GT0090.

6. Remarks: nil.

PO 005 - MANAGE THE OPERATION OF HULL SYSTEMS

1. Performance: Manage the Operation of Hull Systems

2. Conditions:

   a. Given:

      (1) Operational Ship;

      (2) References;

      (3) Engineering Staff Support; and

      (4) Integrated Logistics Support (ILS);.

   b. Denied: N/A.

   c. Environment:

      (1) All Conditions, day or night.

3. Standard: The MS ENG shall manage the operation of hull systems using a total ship's systems integration approach:

   a. IAW:

      (1) CFTO's;

      (2) NEM;

      (3) SSOs; and

      (4) Relevant engineering directives.

4. Reference:

   a. TBD.

5. Task Numbers: OT0155, OT0156, DT0070, DT0065, GT0055, GT0060, DT0050, GT0075, GT0090, QT0025, JT0035, JT0040 and QT0020.

6. Remarks: nil.

PO 006 - DIRECT EMERGENCY OPERATIONS

1. Performance: Direct Emergency Operations

2. Conditions:

   a. Given:

      (1) Operational Ship;

      (2) Supervision;

      (3) References;

      (4) Engineering Staff Support; and

      (5) Integrated Logistics Support (ILS).

   b. Denied: N/A.

   c. Environment:

      (1) All Conditions, day or night.

3. Standard: Under supervision the MS ENG shall direct emergency operations.

   a. IAW:

      (1) CFTO's;

      (2) NEM;

      (3) SSOs;

      (4) CFCD 102;

      (5) Ship's Class Engineering Drills Manual; and

      (6) Relevant engineering directives.

4. Reference:

   a. TBD.

5. Task Numbers: ET0030, ET0035, ET0005 and CT0015.

6. Remarks: nil.

PO 007 - MANAGE SHIP'S STABILITY

1. Performance: Manage Ship's Stability
2. Conditions:
    a. Given:
        (1) Operational Ship;
        (2) Supervision
        (3) References;
        (4) Engineering Staff Support; and
        (5) Stability Assessment Software.
    b. Denied: N/A.
    c. Environment:
        (1) All Conditions, day or night.
3. Standard: Under supervision the MS ENG shall manage ship's stability:
    a. IAW:
        (1) Ship's Manual of Trim and Stability;
        (2) NAVORDs;
        (3) Ship's Class Environnemental Management System (EMS) Manual;
        (4) Relevant engineering directives.
4. Reference:
    a. TBD.
5. Task Numbers: PT0005, PT0020 and PT0015.
6. Remarks: nil.

PO 008 - DIRECT DAMAGE CONTROL OPERATIONS

1. Performance: Direct Damage Control Operations

2. Conditions:

   a. Given:

      (1) Operational Ship;

      (2) Supervision;

      (3) References;

      (4) Engineering Staff Support; and

      (5) Stability Assessment Software;.

   b. Denied: N/A.

   c. Environment:

      (1) All Conditions, day or night.

3. Standard: Under supervision the MS ENG shall direct damage control operations:

   a. IAW:

      (1) Damage Control Manual;

      (2) Ship's Manual of Trim and Stability;

      (3) NEM;

      (4) SSOs;

      (5) CFCD 102; and

      (6) Relevant engineering directives.

4. Reference:

   a. TBD.

5. Task Numbers: ET0025, ET0010, CT0010, ET0015, DT0085 and ET0020.

6. Remarks: nil.

PO 009 - MANAGE SHIPBOARD LIQUIDS

1. Performance: Manage Shipboard Liquids

2. Conditions:

    a. Given:

        (1) Operational Ship;

        (2) Supervision

        (3) References;

        (4) Engineering Staff Support;

        (5) Integrated Logistics Support (ILS); and

        (6) Stability Assessment Software.

    b. Denied: N/A.

    c. Environment:

        (1) All Conditions, day or night.

3. Standard: Under supervision the MS ENG shall manage shipboard liquids:

    a. IAW:

        (1) Ship’s Manual of Trim and Stability

        (2) CFTO’s;

        (3) NEM;

        (4) SSOs; NAVORDs;

        (5) Ship’s Class Environnemental Management System (EMS) Manual; and

        (6) Relevant Harbour Control Instructions (HCI).

4. Reference:

    a. TBD.

5. Task Numbers: PT0020, PT0030, PT0040, PT0035, PT0010, DT0100, PT0055, PT0050, PT0060 and PT0045.

6. Remarks: nil.

PO 010 - MANAGE DOCKING OPERATIONS

1. Performance: Manage Docking Operations

2. Conditions:

   a. Given:

      (1) Operational Ship;

      (2) Supervision;

      (3) References;

      (4) Engineering Staff Support;

      (5) Stability Assessment Sofware; and

      (6) Docking Facilities;

   b. Denied: N/A.

   c. Environment:

      (1) All Conditions, day or night.

3. Standard: Under supervision the MS ENG shall manage docking operations.

   a. IAW:

      (1) Captain’s Ship Book;

      (2) Ship’s Manual of Trim and Stability;

      (3) CFTO’s;

      (4) NEM;

      (5) NaMMs;

      (6) Ship’s Class Environmental Management System (SEMs) Manual;

      (7) Relevent Harbour Control Instructions (HCI); and

      (8) Relevant engineering directives.

4. Reference:

a. TBD.

5. Task Numbers: IT0020, IT0025, IT0015, DT0090, IT0030 and IT0035.

6. Remarks: nil.

PO 011 - ADMINISTER RISK MANAGEMENT PROGRAMS

1. Performance: Administer Risk Management Programs

2. Conditions:

   a. Given:

      (1) Operational Ship;

      (2) References;

      (3) Engineering Staff Support;

      (4) Formation Safety Staff Support;

      (5) Fire Reporting Software; and

      (6) Integrated Logistics Support (ILS);

   b. Denied: N/A.

   c. Environment:

      (1) All Conditions, day or night.

3. Standard: The MS ENG shall administer risk management programs:

   a. IAW:

      (1) DAOD's;

      (2) NAVORDs

      (3) NEM;

      (4) Ship's Class Environmental Management System (SEMs) Manual; and

      (5) Canadian Forces General Safety Program and Canada Labour Code Part II.

4. Reference:

   a. TBD.

5. Task Numbers: FT0005, FT0015, AT0040, AT0045, FT0025, FT0020 FT0040 and FT0045.

6. Remarks: nil.

TRAINING PLAN

CHAPTER 3 – TRAINING MANAGEMENT DETAILS

TRAINING STRATEGY

1. The following training strategies were selected for this course:

   a. Marine Systems Engineering Application Course (MSEAC), which is the ashore phase of training. MSEAC will be delivered through a series of lectures, labs and equipment tours; and

   b. The afloat phase of training which require self-study and mentorship from the MSE department of the trainee's ship.

TRAINING DURATION

2. Based on existing resources and approved instructional design requirements, the approved Course Design authorizes 110 training days.

REQUIRED COURSE STAFFING

3. The following key personnel are required to conduct the training.

   a. This information was not found in the previous QSP. This will be filled out at a later date.

Course Director				
Rank	MOSID	Qualifications Required	Qty	PO / EO

Instructors				
Rank	MOSID	Qualifications Required	Qty	PO / EO

Guest Lecturers/Others (May be civilian)				
Lecturer	# Instructor	Subject	EO #	# period

TRAINING LIMITATION

4. A training limitation may be due to constraints of resources, personnel or equipment. A training limitation is the difference between what a qualified individual must be able to do and what training can prepare them to do. The following limitations are identified below:

a. Nil

COURSE CAPACITY

5. The capacity per session is based on the guidelines in CFITES Vol 5 – Instructor to Learner Ratios:

a. Optimum – 15;

b. Maximum Load – 24; and

c. Minimum – 6.

DESIGNATED SCHEDULING AND COURSE LOADING

6. Training will be scheduled utilizing the HRMS Training (MITE) module to meet the training requirements (IAW NPTORD 4540-2).

7. The authority to load members onto this course will be IAW NPTORD 4540-43. NPTG HQ is to be included as an info addressee on all loading messages. Course loading will be accomplished by:

a. The Career Manager.

QUALIFICATION AND ASSESSMENT

8. This is a pass/fail course. Students must successfully complete all POs to achieve the qualification.

9. Students will be monitored and assessed both on performance and conduct throughout the course. Counselling, progress monitoring, and maintenance of student files will be in accordance with NPTORD 4500-51. Although not mandated specifically in the POs, the general demands of service members and the leadership demands of junior and senior leaders form an integral part of individual training. As such, students are presumed to possess the leader and/or follower attributes commensurate with their rank.

UNSATISFACTORY PROGRESS

10. Any student who fails to successfully complete all POs in the allocated number of attempts will be referred to the Commandant (Cmdt) or Commanding Officer (CO) of the training establishment (TE) to determine the requirement for a TRB IAW NPTORD 4500-54.

REMOVAL FROM TRAINING

11. A TE CO/Cmdt may direct that a student be removed from training for reasons of deficiency in performance, conduct (discipline, attitude, teamwork, safety, leadership), missed training, medical, and other non-performance/conduct related issues. In these

cases, Cease Training Procedures are to be strictly adhered to IAW NPTORD 4500-29 or 4500-54.

RECERTIFICATION REQUIREMENTS

12. There are no additional requirements to maintain currency in this qualification; or Currency.

LANGUAGE OF INSTRUCTION

13. The language of instruction shall be in accordance with the Delivery of Training and Education in Both Official Languages (DAOD 5039-6).

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

14. PLARs shall be conducted in accordance with DAOD 5031-1 Canadian Forces Military Equivalencies Program, through the RCN PLAR submissions coordinator.

POST COURSE REVIEWS (PCR)

15. PCRs are to be conducted on all steady state courses IAW NPTORD 4500-23.

COURSE REPORTING

16. Formal Course: A Canadian Forces Course Report, Form CF 377 upon completion of the ashore phase and CF289 Certificate of Achievement upon successful completion of Qualification Board, shall be prepared and distributed in accordance with DAOD 5031-9.The Course Report shall be descriptive, identifying the student's strengths and weaknesses with specific comments on any PO for which the student demonstrated outstanding performance or had difficulty achieving the standard, any conduct relevant to course performance, and any conditions or situations that affected course performance.

TRAINING ESTABLISHMENTS

17. The designated TE(s) for this training can be found in annex N.

CHAPTER 4 LESSON SPECIFICATIONS

PO 001 – MANAGE THE OPERATION OF MAIN PROPULSION SYSTEMS

EO 001.01 DESCRIBING THE THEORY AND APPLICATION OF MARINE GAS TURBINE

1. Performance: Describe the Theory and Application of Marine Gas Turbines.

2. Conditions:

   a. Given:

      (1) Supervision; and

      (2) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) N/A.

2. Standard: The MS ENG shall demonstrate a thorough understanding of the theory and application of Marine Gas Turbines.

3. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Definition and Purpose**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
2	**History and Development**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
3	**Advantages and Disadvantages**	**Method:** Lecture **Media:** PowerPoint		

Teaching Points		Instructional Strategies	Time (min)	References
		Environment: Classroom		
4	**Theoretical Overview and Terminology**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
4.1	Brayton cycle			
4.2	Comparison with Rankine and Otto cycles;			
4.3	Efficiency; and			
4.4	Compressor theory: 1. Requirements; 2. Types (Axial vs. Centrifugal); 3. Stage theory; 4. Surge and stall; and 5. Variable Guide vanes.			
4.5	Turbine theory: 1. Requirements; and 2. Types (reaction vs. impulse).			
4.6	Combustion theory: 1. Combustion requirements; 2. Types of combustion systems; 3. Requirements of burners; and 4. Types of burners.			
5	**Components**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
5.1	1. Intake; 2. LP compressor; 3. Auxiliary drives; 4. HP compressor; 5. Combustor and nozzles; 6. Turbine section; 7. Power turbine;			

Teaching Points		Instructional Strategies	Time (min)	References
	8. Exhaust; 9. Spooling and bearings; 10. Sealing arrangements; 11. Air Bleeds; and 12. Auxiliaries: i. Fuel control systems; ii. Lubricating oil systems; iii. Starting systems; and iv. Ignition systems.			
6	**Gas Turbine Materials:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
6.1	Material requirements;			
6.2	Types; and			
6.3	Coatings.			
7	**Operational Considerations:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
7.1	SOA;			
7.2	Engine selection;			
7.3	Engine/shaft combinations;			
7.4	Economical running; and			
7.5	Fuel consumption.			
8	**Starting and Stopping:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
8.1	Pre-start checks;			
8.2	Start permissives;			
8.3	Auto start sequence; and			
8.4	Starts: 1. Manual; and			

Teaching Points		Instructional Strategies	Time (min)	References
	2. Emergency.			
8.5	Stopping: 1. Normal; and 2. Emergency.			
9	**Running Parameters and Protection:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
9.1	Parameters monitored;			
9.2	Warnings, alarms and trips; and			
9.3	Fire detection and suppression.			
10	**Gas Turbine Trials;**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
11	**Installation and Removal Considerations:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
11.1	Intakes;			
11.2	Filtration;			
11.3	Silencing;			
11.4	Mounting;			
11.5	Acoustic enclosure;			
11.6	Uptakes;			
11.7	IR suppression;			
11.8	Enclosure air; and			
11.9	GT removal routes.			
12	**Troubleshooting and Maintenance Considerations:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		

Teaching Points		Instructional Strategies	Time (min)	References
12.1	Gas Path Analysis;			
12.2	Hot Section Inspection;			
12.3	Water and Detergent washing; and			
12.4	Dry motoring			
13	**Corrosion and Preservation**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
14	**Safety and Environmental Considerations**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
15	**Administration**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
15.1	Gas Turbine Logs			
16	**In-Service Gas Turbines**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
16.1	GE LM2500			
16.2	RN; and			
16.3	USN			
17	**The practical instruction shall include the following activities:**	**Method:** OJT **Media:** N/A **Environment:** Ship		
17.1	Component and auxiliary systems identification			
17.2	Flash-up			
17.3	Rounds			
17.4	Monitoring Gas Turbine running parameters			
17.5	Routine running maintenance			

Teaching Points		Instructional Strategies	Time (min)	References
17.6	Troubleshooting (using controlled induced faults)			
17.7	Shutdown; and			
17.8	Borescope/Videoscope			
18	**Present marine gas turbines to include**	**Method:** OJT **Media:** N/A **Environment:** Ship		
18.1	Intake and Uptakes			
18.2	Gas Generator and power Turbine			
18.3	Fuel System			
18.4	Lubrication system			
18.5	Engine mounting arrangements			
19	**System walkthrough to cover the following**	**Method:** OJT **Media:** N/A **Environment:** Ship		
19.1	Intake and Uptakes			
19.2	Gas Generator and power Turbine			
19.3	Fuel System			
19.4	Lubrication system			
19.5	Engine mounting arrangements			
19.6	Accessory gearbox			
19.7	Hydraulic start system			
20	**Observe and Participate in marine gas turbine operations to include**	**Method:** OJT **Media:** N/A **Environment:** Ship		
20.1	Flash-up and Shut down			
20.2	Dry Motoring			
20.3	Periodic maintenance			
20.4	LOP Operations			
20.5	Main Machinery Performance Trial (MMPT)			

4. Total EO Time: Total: 44 x 50 minute periods as follows:

a. 21 x 50 min period – Classroom;

b. 21 x 50 min period – Practical; and

c. 2 x 50 min periods – EC (indicate total time or per student).

5. Training and Learning Aids:

a. Static Marine Gas Turbine Trainer;

b. IPMS Trainer; and

c. Gas Turbine simulation software

6. Remarks: Nil.

PO 001 – MANAGE THE OPERATION OF MAIN PROPULSION SYSTEMS

EO 001.02 DESCRIBE THE THEORY AND APPLICATION OF PROPULSION DIESELS

1. Performance: Describe the Theory and Application of Propulsion Diesels.

2. Conditions:

   a. Given:

      (1) Supervision; and

      (2) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) N/A.

3. Standard: The MS ENG shall demonstrate a thorough understanding of the theory and application of Propulsion Diesels.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Definition and Purpose**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
2	**History and Development**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
3	**Advantages and Disadvantages**	**Method:** Lecture **Media:** Powerpoint		

Teaching Points		Instructional Strategies	Time (min)	References
		Environment: Classroom		
4	**Classification of diesel engines**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
5	**Theoretical Overview and Terminology**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
5.1	Otto cycle			
5.2	Diesel cycle			
5.3	Dual cycle			
5.4	Piston speed			
5.5	Bore/stroke			
5.6	Engine speed			
5.7	Compression ratio			
5.8	Maximum cylinder pressure			
5.9	Space and weight constraints			
5.10	Efficiency			
5.11	Combustion theory: 1. Compression ignition requirements; 2. Fuel/air ratio; 3. Phases of combustion; 4. Advanced and delayed injection; and 5. Air flow (swirl and squish)			
5.12	Scavenging: 1. 2-stroke scavenge; 2. 2-stroke timing; 3. 4-stroke scavenge; and 4. 4-stroke timing			
6	**Components:**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		

Teaching Points		Instructional Strategies	Time (min)	References
6.1	Turbo charging and super charging: 1. Advantages and effects; 2. Types of turbo chargers; 3. Types of super chargers; 4. Turbo charger lag; 5. Lubrication; 6. Mounting arrangements; and 7. Boost pressure.			
6.2	Engine block and crankcase: 1. Function; 2. Types; 3. Construction; and 4. Crankcase high pressure			
6.3	Crankshafts: 1. Function; 2. Construction; 3. Materials; 4. Counter weights; 5. Flywheel; 6. Auxiliary drives; 7. Torsional vibration; and 8. Crankshaft deflections			
6.4	Connecting rods and bearings: 1. Function; 2. Configurations; 3. Bearing types and materials; 4. Bearing lubrication; and 5. Failure modes			
6.5	Pistons and piston rings: 1. Function; 2. Piston materials; 3. Types of piston rings; and 4. Lubrication			
6.6	Cylinder heads: 1. Function; 2. Construction; 3. Materials; and			

Teaching Points		Instructional Strategies	Time (min)	References
	4. Test cocks.			
6.7	Valves: 1. Function; 2. Materials; 3. Operation; and 4. Cam shaft			
6.8	Cylinder liners: 1. Function; 2. Types; 3. Scaling; and 4. Erosion and corrosion.			
6.9	Engine mounting arrangements and enclosures: 1. Function and requirements; and 2. Types.			
6.10	Fuel systems: 1. Requirements; 2. Types; 3. Pumps; 4. Injectors; and 5. Filtration			
6.11	Cooling systems: 1. Function; and 2. Types.			
6.12	Lubricating oil systems: 1. Function; 2. Configurations; 3. Lubricating oil characteristics; and 4. Filtration			
6.13	Starting systems: 1. Function; and 2. Types and configurations.			
6.14	Governors: 1. Function and requirements; and			

Teaching Points		Instructional Strategies	Time (min)	References
	2. Types and operation.			
6.15	Intakes and exhaust: 1. Requirements; 2. Filtration; 3. Noise reduction; 4. Exhaust Gas Treatment & Selective Catalytic Reduction 5. IR Suppression; and 6. Intercoolers.			
7	**Operational Considerations:**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
7.1	SOA;			
7.2	Economical Running			
7.3	Fuel Consumption			
8	**Starting and Stopping:**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
8.1	Pre-start checks;			
8.2	Start permissives;			
8.3	Auto start sequence; and			
8.4	Abnormal Starts: 1. Manual; and 2. Emergency.			
8.5	Stopping: 1. Normal; and 2. Emergency.			
9	**Running Parameters and Protection:**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
9.1	Parameters monitored;			

Teaching Points		Instructional Strategies	Time (min)	References
9.2	Warnings, alarms and trips; and			
9.3	Fire detection and suppression.			
10	**Diesel Trials;**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
11	**Installation and Removal Considerations:**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
11.1	Intakes;			
11.2	Filtration;			
11.3	Silencing;			
11.4	Mounting;			
11.5	Acoustic enclosure;			
11.6	Exhaust			
11.7	IR suppression;			
12	**Troubleshooting and Maintenance Considerations:**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
12.1	Diesel Inspections			
12.2	Diesel trap;			
12.3	Overhauls			
12.4	Water Washing			
12.5	Common faults in CF diesel engines			
12.6	Preservation			
12.7	Coking/Decoking. 1. Purpose; 2. Policy; and 3. Procedure			
13	**Corrosion and Preservation**	**Method:** Lecture		

Teaching Points		Instructional Strategies	Time (min)	References
		Media: Powerpoint **Environment:** Classroom		
14	**Safety and Environmental Considerations**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
15	**Administration**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
16	**In-Service Diesels**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
16.1	Pielstick 20PA6 V 280			
16.2	Caterpillar C32			
16.3	MWM D226-6			
16.4	Wärtsilä SACM UD23			
16.5	GM			
16.6	Valenta 16RPA200SZ			
16.7	MAN (AOPV Main Engines)			
16.8	Caterpillar (AOPV Emergency engine)			
16.9	MAN 12V32/44CR			
16.10	RN			
16.11	USN			
17	**The practical instruction shall include the following activities:**	**Method:** OJT **Media:** N/A **Environment:** Ship		
17.1	Component and auxiliary systems identification			
17.2	Flash-up			
17.3	Rounds			
17.4	Monitoring Diesel running parameters			

Teaching Points		Instructional Strategies	Time (min)	References
17.5	Routine running maintenance			
17.6	Troubleshooting (using controlled induced faults)			
17.7	Shutdown; and			
17.8	Borescope/Videoscope			
17.9	Diesel Trap			

5. Total EO Time: Total: 44 x 50 minute periods as follows:

   a. 21 x 50 min period – Classroom;

   b. 21 x 50 min period – Practical; and

   c. 2 x 50 min periods – EC (indicate total time or per student).

6. Training and Learning Aids:

   a. Operational diesel test cell.

7. Remarks: Nil.

PO 001 – MANAGE THE OPERATION OF MAIN PROPULSION SYSTEMS

EO 001.03 DESCRIBE THE THEORY AND APPLICATION OF STEAM PROPULSION SYSTEMS

1. Performance: Describe the Theory and Application of Steam Propulsion Systems.

2. Conditions:

   a. Given:

      (1) Supervision; and

      (2) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) N/A.

3. Standard: The MS ENG shall demonstrate a thorough understanding of the theory and application of Steam Propulsion Systems.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Definition and Purpose**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
2	**History and Development**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
3	**Advantages and Disadvantages**	**Method:** Lecture **Media:** Powerpoint		

Teaching Points		Instructional Strategies	Time (min)	References
		Environment: Classroom		
4	**Theoretical Overview and Terminology**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
4.1	Rankine cycle			
4.2	Turbine theory 1. Requirements; and Types (reaction vs. Impulse			
4.3	Space and weight constraints			
4.4	Efficiency			
5	**Components:**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
5.1	Boilers: 1. Steam drums; 2. Water drum; 3. Headers; 4. Downcomers; 5. Steam generating, water walls and firewall tubes; 6. Superheater; 7. Control de-superheater; 8. Economizer; 9. Forced draught fans; 10. Casing; 11. Furnace and brickwork; 12. Burner register; 13. Soot blowers; 14. Safety valves (pilot and unloader); 15. Main stop valve and by-pass valve; and 16. Control systems			
5.2	Turbines (HP and LP): 1. Astern turbine; 2. Turbine rotor and stator; 3. Turbine casings;			

Teaching Points		Instructional Strategies	Time (min)	References
	4. Gland sealing arrangements; 5. Journal and thrust bearings; 6. Clearances; 7. Expansion arrangements; 8. Main condensers 9. Turbine drains; 10. Engine/transmission coupling arrangements; 11. Steam chest; and 12. Main engine throttle valves			
5.3	Closed Feed System: 1. De-aerator; 2. Feed pumps; and 3. Feed tanks.			
5.4	Main Steam Range			
5.5	Auxiliaries: 1. Fuel control systems; and 2. Lubricating oil systems			
6	**Operational Considerations:**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
6.1	SOA;			
6.2	Normalizing			
6.3	Prolonged running astern			
6.4	Economical Running			
6.5	Fuel Consumption			
7	**Flash-up and Shutdown:**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
8	**Full Power Parameters and Protection:**	**Method:** Lecture **Media:** Powerpoint		

Teaching Points		Instructional Strategies	Time (min)	References
		Environment: Classroom		
9	**Troubleshooting and Maintenance Considerations:**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
10	**Corrosion and Preservation**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
11	**Safety and Environmental Considerations**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
12	**Administration**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
12.1	Boiler and Engine Logs			
13	**In-Service Steam Propulsion Plant**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		

5. Total EO Time: Total: 11 x 50 minute periods as follows:

   a. 11 x 50 min period – Classroom.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 001 – MANAGE THE OPERATION OF MAIN PROPULSION SYSTEMS

EO 001.04 DESCRIBE THE THEORY AND APPLICATION OF NUCLEAR PROPULSION SYSTEMS

1. Performance: Describe the Theory and Application of Propulsion Diesels.

2. Conditions:

   a. Given:

      (1) Supervision; and

      (2) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) N/A.

3. Standard: The MS ENG shall demonstrate a thorough understanding of the theory and application of Nuclear Propulsion Systems.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Definition and Purpose**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
2	**History and Development**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
3	**Advantages and Disadvantages**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		

Teaching Points		Instructional Strategies	Time (min)	References
4	**Theoretical Overview and Terminology**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
4.1	Nuclear Fission Theory			
4.2	Turbine theory: 1. Requirements; 2. and Saturated turbine and steam reheaters.			
4.3	Efficiency			
5	**Components:**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
5.1	Primary Loop: 1. Reactor: i. Core: 1. Enriched Uranium; 2. Uranium Aluminium Oxides; and 3. Fuel Rods vs Fuel Plates. ii. Control Rods; iii. Shut down rods; iv. Emergency core cooling system; and v. Poison Injection System. 2. Coolant Pump; 3. Steam Generator; and 4. Pressurizer			
5.2	Secondary Loop: 1. Steam Generator; 2. Saturated Turbine; 3. Condenser; 4. Feed Pump; and 5. Reheaters			

Teaching Points		Instructional Strategies	Time (min)	References
5.3	Primary and Secondary Containment: 1. Blast Gates.			
5.4	Auxiliaries: 1. Pressurizer Heating System: 2. Operation in harbour. 3. Saturated Steam Driven Auxiliaries			
6	**Materials**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
7	**Troubleshooting and Maintenance Considerations**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
8	**Corrosion and Preservation**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
8.1	Water chemistry: 1. Primary Loop Radioactivity Monitoring.			
9	**Safety and Environmental Considerations**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
9.1	Poison Injection System			
9.2	Emergency Core Cooling System			
9.3	Shutdown System			
9.4	Containment Systems			
9.5	Radiological Systems			

Teaching Points		Instructional Strategies	Time (min)	References
9.6	Nuclear Emergency Response Team (NERT): 1. Nuclear Port Visit Monitoring; and 2. Radionuclide Baseline Monitoring			

5. Total EO Time: Total: 3 x 50 minute periods as follows:

   a. 3 x 50 min period – Classroom.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 001 – MANAGE THE OPERATION OF MAIN PROPULSION SYSTEMS

EO 001.05 DESCRIBE THE THEORY AND APPLICATION OF ELECTRIC PROPULSION SYSTEMS

1. Performance: Describe the Theory and Application of Electric Propulsion Systems.

2. Conditions:

   a. Given:

      (1) Supervision; and

      (2) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) N/A.

3. Standard: The MS ENG shall demonstrate a thorough understanding of the theory and application of electric propulsion systems.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Definition and Purpose**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
2	**History a`nd Development**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
2.1	Fuel cells			
2.2	Permanent magnets			
2.3	High Temperature super conductors			

Teaching Points		Instructional Strategies	Time (min)	References
3	**Advantages and Disadvantages**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
3.1	Noise			
3.2	Space and weight constraints			
3.3	Efficiency			
3.4	Drive configuration flexibility			
4	**Theoretical Overview and Terminology**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
5	**Components:**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
5.1	Power generation 1. Prime mover; and 2. Generator: i. Types; ii. Size; and iii. Location			
5.2	Energy storage: 1. Batteries; and 2. Associated controls			
5.3	Power conversion (AC - DC), (DC - AC): 1. Diode-rectifier; and 2. Pulse width modulators.			
5.4	Power Distribution: 1. Propulsion switchboard; 2. Ship's service switchboard; 3. Ship's cabling; and 4. Interlocks			
5.5	Propulsion motors:			

Teaching Points		Instructional Strategies	Time (min)	References
	1. Induction motors; 2. DC motors; 3. Brush gear; 4. Commutator; and 5. Armature configuration			
5.6	Platform loads: 1. Armament load; and 2. Hotel load			
5.7	Power Control: 1. Thyristors; and 2. Field supply converter			
6	**Operational Considerations:**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
6.1	SOA;			
6.2	Fuel consumption			
6.3	Economical Running			
6.4	Battery Management			
7	**Running Parameters and Protection:**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
7.1	Voltages;			
7.2	Currents			
7.3	Engaging and disengaging main motor			
8	**Troubleshooting and Maintenance Considerations:**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
8.1	Insulation tests;			
8.2	Commutator Wear			
8.3	Brush gear			
9	**Corrosion and Preservation**	**Method:** Lecture		

Teaching Points		Instructional Strategies	Time (min)	References
		Media: Powerpoint **Environment:** Classroom		
10	**Safety and Environmental Considerations**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
10.1	Bonding and Grounding			
10.2	High Voltages			
11	**In-Service Systems**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
11.1	KIN Class			
11.2	HDW Class			
11.3	VIC Class			

5. Total EO Time: Total: 6 x 50 minute periods as follows:

   a. 4 x 50 min period – Classroom; and

   b. 2 x 50 min periods – EC (indicate total time or per student).

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 001 – MANAGE THE OPERATION OF MAIN PROPULSION SYSTEMS

EO 001.06 DESCRIBE THE THEORY AND APPLICATION OF TRANSMISSION SYSTEMS

1. Performance: Describe the Theory and Application of Transmissions Systems.

2. Conditions:

   a. Given:

      (1) Supervision; and

      (2) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) N/A.

3. Standard: The MS ENG shall demonstrate a thorough understanding of the theory and application of transmission systems.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Definition and Purpose**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
1.1	General Requirement for transmission system			
1.2	Naval requirement for transmission system			
2	**History and Development**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
2.1	Gearbox explosions of the past			

Teaching Points		Instructional Strategies	Time (min)	References
3	**Theoretical Overview and Terminology**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
3.1	Torque/Power; and			
3.2	Reduction Ratio			
3.3	Prime mover vs shaft speeds			
3.4	Reverse capability			
3.5	Clutching arrangements			
3.6	Principles of Gearing: 1. Gearing Geometry/Profiles 2. Novikov 3. VBB, Vickers Bostock Bramley 4. Involute			
3.7	Gear Meshing: 1. Meshing 2. Action/Dynamics 3. Pinion Deflections 4. Noise Reduction			
4	**Gearing Construction:**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
4.1	Manufacturing: 1. Hobbing 2. Shaping			
4.2	Tooth Finishing: 1. Grinding 2. Shaving			
4.3	Hardening Process: 1. Gas Carburizing 2. Nitriding 3. Induction Hardening			

Teaching Points		Instructional Strategies	Time (min)	References
4.4	Materials			
5	**Gearing Tooth Failures:**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
5.1	Mechanics of failure			
5.2	Inspection and type of failure: 1. Tooth root fracture 2. Pitting 3. Scuffing 4. Abrasive wear 5. Ridging 6. Corrosive wear			
6	**Gearing Loads:**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
7	**Gearing Arrangements:**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
7.1	Gear Train Arrangements 1. Single or Double Helical 2. Single or Double Reduction 3. Single or Dual Tandem 4. Articulated 5. Locked Train			
7.2	Gearbox Construction and Gearing Arrangements			
8	**Gear Tooth Lubrication:**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
9	**Principles of Bearings:**	**Method:** Lecture		

Teaching Points		Instructional Strategies	Time (min)	References
		Media: Powerpoint **Environment:** Classroom		
9.1	Design and Dynamics of operation: 1. Construction 2. Lubrication i. Functions of Lubrication ii. Principles of Lubrication iii. Selection of Lubricants iv. Temperature Range			
9.2	Bearing types and applications: 1. Journal bearings 2. Ball bearings 3. Roller bearings 4. Thrust bearings 5. Anti-friction bearings			
9.3	Bearing Failure: 1. Condition monitoring: i. Vibration Analysis ii. Temperature iii. Oil analysis iv. Run down time 2. Types of failure i. Fatigue ii. Installation/improper fits iii. Corrosion iv. Contamination v. Lubrication breakdown vi. Loss of lubrication			

Teaching Points		Instructional Strategies	Time (min)	References
10	**Principles of Clutches:**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
10.1	Functions			
10.2	Requirements			
10.3	Types of clutches: 1. Dog clutch 2. Plate clutch: i. Pneumatic clutch ii. Centrifugal clutch 3. Electric clutch 4. SSS (Self-Shifting Synchronization) Clutch			
10.4	Lubrication			
11	**Running Parameters and Protection:**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
11.1	Max Torque Limit			
11.2	Bearing Monitoring			
11.3	Gearbox vibration			
12	**Troubleshooting and Maintenance Considerations:**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
12.1	Gearbox Inspection: 1. Security			
13	**Corrosion Prevention and Preservation Techniques**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
14	**In-Service Systems:**	**Method:** Lecture **Media:** Powerpoint		

Teaching Points		Instructional Strategies	Time (min)	References
		Environment: Classroom		
14.1	HAL Class			
14.2	HDW Class			
14.3	PRO Class			

5. Total EO Time: Total: 22 x 50 minute periods as follows:

   a. 18 x 50 min period – Classroom; and

   b. 4 x 50 min periods – Workshop demonstration.

6. Training and Learning Aids:

   a. Statics displays of gearbox cut-away;

   b. Transmission clutch assembly;

   c. Gearbox bearing; and

   d. Lubricating oil arrangements.

7. Remarks: Nil.

PO 001 – MANAGE THE OPERATION OF MAIN PROPULSION SYSTEMS

EO 001.07 DESCRIBE THE THEORY AND APPLICATION OF MAIN PROPULSION POWER TRAIN

1. Performance: Describe the Theory and Application of Main Propulsion Power Train.

2. Conditions:

   a. Given:

      (1) Supervision; and

      (2) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) N/A.

3. Standard: The MS ENG shall demonstrate a thorough understanding of the theory and application of main propulsion power train.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Shafting**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
1.1	Definition and Purpose			
1.2	Theoretical Overview and Terminology: 1. Length; 2. Torque/torsional flexibility 3. Number of shafts; 4. Alignment; 5. Support; and 6. Rake and spay			

Teaching Points		Instructional Strategies	Time (min)	References
1.3	Construction and material			
1.4	Components: 1. Thrust block; 2. Plummer block; 3. Bulkhead gland; 4. Couplings: i. Flexible; ii. Loose; and iii. Solid. 5. Stern seals; 6. Shaft brake; and 7. Brackets: i. “A”, “P”, and “T”. 8. Rope guard			
1.5	Corrosion preservation			
1.6	Signature Reduction: 1. Noise reduction; 2. Magnetic signature; and 3. Electromagnetic signature			
1.7	Operational Considerations: 1. Min/max shaft speed; 2. Shaft locking; 3. Shaft trailing; 4. Torque limits; and 5. Bollard condition.			
1.8	Installation and Removal Considerations			
2	**Propellers**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
2.1	Definition and Purpose			
2.2	Theoretical Overview and Terminology: 1. Shaft speed vs ship speed; 2. Pitch; 3. Number of blades; 4. Noise; and 5. Control/reversible;			

Teaching Points		Instructional Strategies	Time (min)	References
	6. Rotation direction; and 7. Ship manoeuvrability			
1.3	Construction and material			
1.4	Propeller types, advantages and disadvantages: 1. Fixed; 2. CPP; 3. CRPP; and 4. Vertical Axis			
1.5	Propeller cavitation: 1. Definition and theory; 2. Cause; and 3. Prevention/reduction methods			
1.6	Propeller pitch control types: 1. Wet/dry hub			
1.7	Installation and removal considerations			
4	**Troubleshooting and Maintenance Considerations:**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
4.1	Torsion meter arrangement			
4.2	Poker gauge measurement			
4.3	Turning motors			
4.4	Visual inspections by divers			
4.5	Corrosion/erosion/pitting prevention			
5	**Safety and Environmental Considerations**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
6	**In-Service Systems (with emphasis on propellers):**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		

Teaching Points		Instructional Strategies	Time (min)	References
6.1	HAL Class			
6.2	HDW Class			
6.3	PRO Class			
6.4	KIN Class			
6.5	VIC Class			

5. Total EO Time: Total: 20 x 50 minute periods as follows:

   a. 17 x 50 min period – Classroom; and

   b. 3 x 50 min periods – Workshop demonstration.

6. Training and Learning Aids:

   a. Static displays of a shaft cut-away;

   b. Shaft coupling cut-away (loose and/or flexible);

   c. Shaft block cut-away (plummer and /or thrust);

   d. Bulkhead gland and stern seal cut-away;

   e. A-bracket cut-way;

   f. CRPP propeller; and

   g. Hub cut away

7. Remarks: Nil.

PO 001 – MANAGE THE OPERATION OF MAIN PROPULSION SYSTEMS

EO 001.08 DESCRIBE THE THEORY AND APPLICATION OF MAIN PROPULSION ANCILLARY SYSTEMS

1. Performance: Describe the Theory and Application of Main Propulsion Ancillary systems.

2. Conditions:

   a. Given:

      (1) Supervision; and

      (2) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) N/A.

3. Standard: The MS ENG shall demonstrate a thorough understanding of the theory and application of main propulsion ancillary systems.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Definition and Purpose**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
2	**Types of Ancillaries:**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
2.1	Lubricating oil systems			
2.2	Fuel systems			
2.3	Cooling systems			

Teaching Points		Instructional Strategies	Time (min)	References
3	**Components:**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
3.1	Main lubricating oil 1. Tanks i. Capacity 2. Sumps 3. Pumps/motors i. Flow rate 4. Filters/strainers 5. Coalescers/centrifuges 6. Coolers 7. Supply/return lines 8. Lubricating oil heaters			
3.2	Fuel oil services: 1. Tanks: i. Service; ii. Settling; iii. Emergency/header; and iv. Capacity. 2. Pumps/motors; i. Flow rate. 3. Filters/coalescers; 4. Supply/return lines; and 5. Heaters.			
3.3	Main Sea Water cooling: 1. Sea bays; 2. Strainers; 3. Pumps/motors; 4. Heat exchangers; 5. Temperature controlling valves; 6. Supply/return lines; 7. Cross connect with alternate cooling; and 8. Capacity and flow rate.			

Teaching Points		Instructional Strategies	Time (min)	References
4	**Running Parameters and Protection**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
5	**Troubleshooting and Maintenance Considerations:**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
5.1	Routine filter, heat exchanger and strainer maintenance.			
6	**Corrosion and Preservation**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
7	**Safety and Environmental Considerations**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
8	**In-Service Systems:**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
8.1	HAL Class			
8.2	HDW Class			
8.3	PRO Class			
8.4	VIC Class			

5. Total EO Time: Total: 6 x 50 minute periods as follows:

   a. 4 x 50 min period – Classroom; and

   b. 2 x 50 min periods – EC (indicate total time or per student).

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 001 – MANAGE THE OPERATION OF MAIN PROPULSION SYSTEMS

EO 001.09 DESCRIBE MAIN PROPULSION SYSTEMS IN HMC SHIPS

1. Performance: Describe Main Propulsion Systems in HMC Ships.

2. Conditions:

   a. Given:

      (1) Operational Ship;

      (2) Supervision; and

      (3) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) All conditions, day and night.

3. Standard: The MS ENG shall demonstrate a thorough understanding of the main propulsion systems fitted on board (as applicable).

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Prime Movers**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
1.1	Gas Turbines: 1. Intake and uptake ducting systems and components; 2. Spooling arrangements; 3. Compressors (LP, HP); 4. Gas generator; 5. External air bleeds; 6. Combustion chambers (including igniters and sprayers); 7. Power turbine;			

Teaching Points		Instructional Strategies	Time (min)	References
	8. Air/gas flow; 9. Main engine bearings; 10. Lube oil systems; 11. Fuel systems; 12. Auxiliary drives; 13. Starting system; 14. Engine-transmission coupling arrangement; 15. Engine mounting and expansion arrangement; 16. Safety devices; 17. Engine enclosure; 18. Engine inlet and washing arrangements; and 19. Fire detection and suppression systems			
1.2	Diesels: 1. Intake and exhaust; 2. Lubrication; 3. Fuel system, including pumps and injectors; 4. FW/SW cooling; 5. Crankcase and relief arrangement (explosion doors 6. Wet/dry sump arrangement; 7. Camshaft/crankshaft; 8. Piston and ring arrangement; 9. Engine gearing arrangement; 10. Cylinder arrangement including heads and valves; 11. Connecting rods and bearings; 12. Governor; 13. Auxiliary drives and pumps; 14. Starting system; 15. Turbocharger and washing arrangement; 16. Engine-transmission coupling arrangement;			

Teaching Points		Instructional Strategies	Time (min)	References
	17. Engine mountings; 18. Engine enclosure; 19. Fire detection and suppression systems; and 20. Safety devices			
1.3	Steam Propulsion: 1. Boilers: i. Steam drum and steam drum internals; ii. Water drum; iii. Headers; iv. Downcomers; v. Steam generating, water walls and firewall tubes; vi. Superheater; vii. Control de-superheater; viii. Economizer; ix. Forced draught fans; x. Casing; xi. Furnace and brickwork; xii. Burner register; xiii. Soot blowers; xiv. Safety valves (pilot and unloader); xv. Main stop valve and by-pass valve; and xvi. Bailey Control system (narrative only). 2. Turbines (HP and LP): i. Astern turbine; ii. Turbine rotor and stator; iii. Turbine casings; iv. Gland sealing arrangement; v. Journal and thrust bearings; vi. Clearances;			

Teaching Points		Instructional Strategies	Time (min)	References
	vii. Expansion arrangements; viii. Main condenser; ix. Turbine dra x. Engine/transmission coupling arrangements; xi. Steam chest; and xii. Main engine throttle valves			
2	**Transmissions**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
2.1	Gearbox			
2.2	Gearing arrangements			
2.3	Couplings			
2.4	Bearings			
2.5	Quillshafts			
2.6	Turning gear			
2.7	Clutches			
2.8	Lubrication arrangements			
3	**Power Train**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
3.1	Shafts and couplings			
3.2	Bulkhead glands			
3.3	Thrust blocks			
3.4	Plummer blocks			
3.5	Propeller arrangement			
3.6	Torsion meters			
3.7	Gland seals			
3.8	Stern tubes			
3.9	Stern bearings			
3.10	Support brackets			

Teaching Points		Instructional Strategies	Time (min)	References
3.11	Main shaft locking and rotation arrangements			
3.12	Active/passive shaft grounding			
4	**Ancillary Systems**	**Method:** Lecture **Media:** Powerpoint **Environment:** Classroom		
4.1	Main Lubricating Oil: 1. Tanks; 2. Sumps; 3. Pumps/motors; 4. Filters; 5. Coalescers/centrifuges; 6. Coolers; and 7. Supply/return lines.			
4.2	Fuel Oil Services: 1. Tanks; 2. Pumps/motors; 3. Filters/Coalescer; and 4. Supply/return lines			
4.3	Main Sea Water Cooling: 1. Pumps/motors; 2. Strainers; and 3. Supply/return lines			
4.4	Controllable Pitch Propellers: 1. Tanks; 2. Sumps; 3. Pumps/motors; 4. Filters; 5. Coolers; 6. Sup ply/return lines; 7. Transfer block; 8. Sealing arrangements; and 9. Locking arrangements			
4.5	Main Steam System			
4.6	Closed Feed System			

5. Total EO Time: Total: Throughout the Afloat phase.

6. Training and Learning Aids: Nil.

7. Remarks: The MS ENG shall produce drawing that are:

   a. Neat/legible;

   b. Consistent with ship's fitted drawings; and

   c. Complete, to include as applicable:

      (1) Content and capacities;

      (2) Direction of flow;

      (3) Monitoring arrangements and parameters;

      (4) Control arrangements;

      (5) Valve types; and

      (6) A uniform colour-code.

   d. Consistent with CFTO C-23-007-001/MD-001, Ship's Piping Systems as the standard for symbols.

PO 001 – MANAGE THE OPERATION OF MAIN PROPULSION SYSTEMS

EO 001.10 DESCRIBE THE OPERATION OF MAIN PROPULSION SYSTEMS IN HMC SHIPS.

1. Performance: Describe the Operation of Main Propulsion Systems in HMC Ships.

2. Conditions:

   a. Given:

      (1) Operational Ship;

      (2) Supervision; and

      (3) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) All conditions, day and night.

3. Standard: The MS ENG shall demonstrate a thorough understanding of the operation of the main propulsion system fitted on board.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**The MS ENG shall participate in all modes of operation (including flash up and shutdown) of main propulsion equipment and systems fitted on board:**	**Method:** OJT **Media:** N/A **Environment:** Ship		
1.1	Prime Movers;			
1.2	Transmissions;			
1.3	Power Train; and			
1.4	Ancillaries.			
2	**The MS ENG shall produce written reports on the operation of the main**	**Method:** OJT **Media:** N/A **Environment:** Ship		

Teaching Points		Instructional Strategies	Time (min)	References
	propulsion equipment and systems listed in para 10a to include:			
2.1	Normal operation;			
2.2	Abnormal operations;			
2.3	Emergency operations;			
2.4	Operating parameters;			
2.5	Operational limitations;			
2.6	Monitoring methods;			
2.7	Flash ups;			
2.8	Shutdowns;			
2.9	Safety considerations;			
2.10	Troubleshooting; and			
2.11	Routine running maintenance.			

5. Total EO Time: Total: Throughout the Afloat phase.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 002 – MANAGE THE OPERATION OF AUXILIARY SYSTEMS

EO 002.01 DESCRIBE THE THEORY AND APPLICATION OF GENERAL MARINE ENGINEERING

1. Performance: Describe the Theory and Application of General Marine Engineering.

2. Conditions:

   a. Given:

      (1) Supervision; and

      (2) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) N/A.

3. Standard: The MS ENG shall demonstrate a thorough understanding of the theory and application of General Marine Engineering.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Pumps/Compressors**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
1.1	Purposes			
1.2	Theory			
1.3	Terminology			
1.4	Types			
1.5	Applications			
2	**Valves**	**Method:** Lecture **Media:** PowerPoint		

Teaching Points		Instructional Strategies	Time (min)	References
		Environment: Classroom		
2.1	Purposes			
2.2	Types: 1. Flat; 2. Mitre; 3. Globe; 4. Parallel; and 5. Spherical			
2.3	Applications: 1. Isolation; and 2. Control: i. Regulating; and ii. Relief 3. Packing			
2.4	Mounts: 1. Purpose; 2. Types: i. Rigid; and ii. Flexible. 3. Materials; and 4. Applications: i. Shock mountings; and ii. Vibration mountings			
2.5	Pipe work: 1. Purpose; 2. Materials; 3. Applications; and 4. Pipe hangers			
2.6	Flanges: 1. Purpose; and 2. Types			
2.7	Gaskets: 1. Purpose; and 2. Materials.			
2.8	Lagging: 1. Purpose; 2. Materials; and			

Teaching Points		Instructional Strategies	Time (min)	References
	3. Applications.			
3	**Bearings**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
3.1	Rolling contact bearing identification: 1. Ball bearings; 2. Extra wide deep groove ball bearings; 3. Shields and seals; 4. Retainer and cage material; 5. Lubrication; 6. Date of manufacture; and 7. Roller bearing			
3.2	Causes of failure in rolling contact bearings: 1. General; 2. Procurement and storage; 3. Pre-assembly; 4. Handling; 5. Installation; 6. Operational failure; and 7. Bearing failure investigation			
3.3	Diagnosis of bearing problems			
3.4	Good bearing maintenance practice			
4	**Workshop capabilities (a basic appreciation of ashore and shipboard Marine Engineering workshops):**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
4.1	Machine shop: 1. Lathe; 2. Milling machine; 3. Tools; and 4. Grinders/cutters			

Teaching Points		Instructional Strategies	Time (min)	References
4.2	Shipwright workshop: 1. Wood working; 2. Welding/brazing; 3. Foundry; and 4. Tools.			
4.3	Electrical workshop: 1. Rewinding; 2. Test equipment; and 3. Tools			
4.4	Control workshop: 1. Test equipment; and 2. Tools.			

5. Total EO Time: Total: 14 x 50 minute periods as follows:

   a. 7 x 50 min period – Classroom; and

   b. 7 x 50 min periods – Workshop demonstartion.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 002 – MANAGE THE OPERATION OF AUXILIARY SYSTEMS

EO 002.02 DESCRIBE THE THEORY AND APPLICATION OF VIBRATION ANALYSIS

1. Performance: Describe the Theory and Application of Vibration Analysis.

2. Conditions:

   a. Given:

      (1) Supervision; and

      (2) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) N/A.

3. Standard: The MS ENG shall demonstrate a thorough understanding of the theory and application of vibration analysis.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Vibration Analysis theory:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
1.1	Definition and Purpose;			
1.2	Introduction to Mechanical Vibration: 1. Simple Harmonic Motion; 2. Terminology: i. Frequency; ii. Amplitude; iii. Displacement; iv. Velocity; v. Acceleration; vi. Oscillation:			

Teaching Points		Instructional Strategies	Time (min)	References
	1. Oscillatory motion; and 2. Non-oscillatory motion. vii. Natural Frequency; viii. Forced Vibration Frequency; ix. Resonance; and x. Damping: 1. Damping Factor; 2. Under Damping; 3. Critical Damping; and 4. Over Damping.			
1.3	Vibration Measurement and Instrumentation: 1. Transducers: i. Capacity probes; ii. Electromagnetic probes; and iii. Accelerometers. 2. Data-Trap.			
2	**Discrete Frequency Analysis;**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
3	**Wear Debris Analysis:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
3.1	Theory of Wear;			
3.2	Wear Debris Analysis Techniques: 1. Physical Wear Debris Analysis: i. Filter Debris Analysis; ii. Magnetic Chip Detector; and iii. Spectrometric Oil Analysis:			

Teaching Points		Instructional Strategies	Time (min)	References
	1. OCCAP.			
4	**Operating Analysis Techniques:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
4.1	Fitted Systems;			
4.2	Shaft breakaway;			
4.3	Rundown time;			
4.4	Vibration Monitoring; and			
4.5	Thermography.			
5	**VA Program**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
5.1	Data collection;			
5.2	Analysis; and			
5.3	Department Training Requirements.			
6	**Agencies:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
6.1	FTA;			
6.2	NETE; and			
6.3	DRDC (A/P) Dockyard Lab.			
7	**Causing of Vibration in Machines:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
7.1	Unbalance;			
7.2	Eccentricity;			
7.3	Misalignment;			
7.4	Mechanical;			
7.5	Damaged/defective Bearings or Gearings;			

Teaching Points		Instructional Strategies	Time (min)	References
7.6	Electromagnetic forces;			
7.7	Hydraulic/aerodynamic forces; and			
7.8	Reciprocating forces;			
7.9	Component/system resonance;			
7.10	Rubbing;			
7.11	Drive belts; and			
7.12	Torque variations.			

5. Total EO Time: Total: 11 x 50 minute periods as follows:

   a. 7 x 50 min period – Classroom;

   b. 3 x 50 min period – Lab Practical; and

   c. 1 x 50 min periods – EC (indicate total time or per student).

6. Training and Learning Aids:

   a. VA equipment.

7. Remarks: Nil.

PO 002 – MANAGE THE OPERATION OF AUXILIARY SYSTEMS

EO 002.03 DESCRIBE THE THEORY AND APPLICATION OF REFRIGERATION SYSTEMS

1. Performance: Describe the Theory and Application of Refrigeration Systems.

2. Conditions:

   a. Given:

      (1) Supervision; and

      (2) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) N/A.

3. Standard: The MS ENG shall demonstrate a thorough understanding of the theory and application of Refrigeration Systems.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Definition and Purpose**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
2	**History and Development**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
2.1	Types of refrigerants, their properties, advantages and disadvantages: 1. Water; 2. Ammonia; 3. Freon; and			

Teaching Points		Instructional Strategies	Time (min)	References
	4. R134A & newer refrigerants			
3	**Theoretical Overview and Terminology**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
3.1	Refrigeration cycle			
4	**Components:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
4.1	Compressors: 1. Reciprocating; 2. Screw; 3. Centrifugal; and 4. Capacity			
4.2	Condensers			
4.3	Thermo Expansion Valve (TEV)			
4.4	Evaporators/chilled water cooler			
4.5	Purge units			
4.6	Dryers			
4.7	Oil/refrigerant separators			
4.8	Sight glasses			
4.9	Super heater/sub-cooler			
5	**Materials**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
5.1	Compatibility			
6	**Starting and Stopping**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
6.1	Refrigeration			
6.2	Chilled water system			

Teaching Points		Instructional Strategies	Time (min)	References
7	**Troubleshooting and Maintenance Considerations**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
7.1	Technician certification			
7.2	Leak detection			
7.3	Purging/recovery			
7.4	Hot gas defrost			
7.5	Routine filter, heat exchanger and strainer maintenance			
7.6	TEV			
8	**Corrosion and Preservation**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
9	**Safety and Environmental Considerations**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
10	**Administration**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
11	**In-Service Systems**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
11.1	HAL Class			
11.2	HDW Class			
11.3	PRO Class			

5. Total EO Time: Total: 12 50 minute periods as follows:

   a. 6 x 50 min period – Classroom; and

   b. 6 x 50 min periods – Lab Practical.

6. Training and Learning Aids:

a. Refrigeration Test Cell.

7. Remarks: Nil.

PO 002 – MANAGE THE OPERATION OF AUXILIARY SYSTEMS

EO 002.04 DESCRIBE THE THEORY AND APPLICATION OF COMPRESSED AIR SYSTEMS

1. Performance: Describe the Theory and Application of Compressed Air Systems.

2. Conditions:

   a. Given:

      (1) Supervision; and

      (2) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) N/A.

3. Standard: The MS ENG shall demonstrate a thorough understanding of the theory and application of Compressed Air Systems.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Definition and Purpose:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
1.1	HP;			
1.2	LP; and			
1.3	Control air.			
2	**Components:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
2.1	Compressors;			
2.2	Dryers/filters;			

Teaching Points		Instructional Strategies	Time (min)	References
2.3	Air/oil separators;			
2.4	Sight glasses;			
2.5	Accumulators and cylinders: 1. Capacity.			
2.6	Pressure reducers; and			
2.7	HP/LP system cross connect.			
3	**Materials:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
3.1	Compatibility.			
4	**Starting and Stopping:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
4.1	Lead/lag; and			
4.2	Start/stop cycle.			
5	**Running Parameters and Protection**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
6	**Troubleshooting and Maintenance Considerations:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
6.1	Routine dryer blow down;			
6.2	Silica medium regeneration;			
6.3	Moisture and contamination;			
6.4	Hydrostatic testing; and			
6.5	System integrity testing.			
7	**Corrosion and Preservation**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		

Teaching Points		Instructional Strategies	Time (min)	References
8	**Safety Considerations:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
8.1	Dieseling; and			
8.2	Personnel hazards. 1. Diving Air quality/testing.			
9	**In-Service Systems**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
9.1	HAL Class;			
9.2	HDW Class; and			
9.3	PRO Class.			

5. Total EO Time: Total: 4 x 50 minute periods as follows:

   a. 2 x 50 min period -- Classroom;

   b. 1 x 50 min period – Workshop demostration; and

   c. 1 x 50 min periods – EC (indicate total time or per student).

6. Training and Learning Aids:

   a. Static display of HP or LP Air compressor.

7. Remarks: Nil.

PO 002 – MANAGE THE OPERATION OF AUXILIARY SYSTEMS

EO 002.05 DESCRIBE THE THEORY AND APPLICATION OF STEERING SYSTEMS

1. Performance: Describe the Theory and Application of Steering Systems.

2. Conditions:

   a. Given:

      (1) Supervision; and

      (2) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) N/A.

3. Standard: The MS ENG shall demonstrate a thorough understanding of the theory and application of Steering Systems.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Definition and Purpose**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
2	**Theoretical Overview and Terminology**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
2.1	Hydraulics			
2.2	Fluid dynamics			
2.3	Ship profile			
2.4	Redundancy			
2.5	Manoeuvrability			
3	**Components**	**Method:** Lecture		

Teaching Points		Instructional Strategies	Time (min)	References
		Media: PowerPoint **Environment:** Classroom		
3.1	Motors			
3.2	Pumps			
3.3	Controls; 1. Consoles: i. Bridge; ii. Secondary; and iii. Local. 2. Amplifiers; 3. Signal Leads; 4. Selector switches; 5. Motors; 6. Controllers; 7. Solenoids; 8. Indicators; 9. Trick wheel; and 10. Control feedback			
3.4	Storage tanks			
3.5	Coolers			
3.6	Actuators: 1. RAM; and 2. Rotary vane actuator			
3.7	Rudder			
3.8	Locking arrangement			
3.9	Accumulators			
4	**Operational Considerations**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
4.1	Redundancy: 1. Steering positions; 2. Power supplies; 3. Hydraulics; 4. Controls; and 5. Cooling			
5	**Running Parameters and Protection**	**Method:** Lecture **Media:** PowerPoint		

Teaching Points		Instructional Strategies	Time (min)	References
		Environment: Classroom		
6	**Steering Tests and Trials**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
6.1	Steering Agreement Test			
6.2	Creep Test (PRO Class only)			
7	**Troubleshooting and Maintenance Considerations**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
7.1	Contamination: 1. Salt water; 2. Air; and 3. Solid			
8	**Corrosion and Preservation**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
9	**Safety and Environmental Considerations**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
10	**In-Service Systems**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
10.1	HAL Class (Wagner & Brown Brothers)			
10.2	HDW Class			
10.3	PRO Class			
10.4	VIC Class			

5. Total EO Time: Total: 18 x 50 minute periods as follows:

   a. 7 x 50 min period – Classroom;

b. 7 x 50 min period – Hydraulic lab practical;

c. 3 x 50 min period – Test cell practical; and

d. 1 x 50 min periods – EC (indicate total time or per student).

6. Training and Learning Aids:

a. Hydraulic lab and operational steering system.

7. Remarks:

a. Hydraulic lab practical: reinforcement of hydraulic principles; and

b. Steering systme test cell practical: Flash-up, shutdown, monitoring of ruinning parameters and toubleshooting of induced faults.

PO 002 – MANAGE THE OPERATION OF AUXILIARY SYSTEMS

EO 002.06 DESCRIBE THE THEORY AND APPLICATION OF AUXILIARY SEA WATER SERVICE SYSTEMS

1. Performance: Describe the Theory and Application of Auxiliary Sea Water Service Systems.

2. Conditions:

   a. Given:

      (1) Supervision; and

      (2) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) N/A.

3. Standard: The MS ENG shall demonstrate a thorough understanding of the theory and application of Auxiliary Sea Water Service Systems.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Definition and Purpose:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
2	**Components:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
2.1	Pumps/motors; 1. Flow rate; and 2. Capacity.			
2.2	Strainers;			

Teaching Points		Instructional Strategies	Time (min)	References
2.3	Coolers;			
2.4	Supply/return lines;			
2.5	Cross connect; and			
2.6	Sea bay.			
3	**Materials:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
3.1	Compatibility.			
4	**Troubleshooting and Maintenance Considerations:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
4.1	Routine filter, heat exchanger and strainer maintenance.			
4.2	Cathelco; and			
4.3	Sacrificial anodes.			
5	**In-Service Systems**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
5.1	HAL Class;			
5.2	HDW Class (seawater and freshwater systems); and			
5.3	PRO Class.			

5. Total EO Time: Total: 1 x 50 minute periods as follows:

   a. 1 x 50 min period – Classroom.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 002 – MANAGE THE OPERATION OF AUXILIARY SYSTEMS

EO 002.07 DESCRIBE THE THEORY AND APPLICATION OF FRESH WATER PRODUCTION SYSTEMS

1. Performance: Describe the Theory and Application of Fresh Water Production Systems.

2. Conditions:

   a. Given:

      (1) Supervision; and

      (2) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) N/A.

3. Standard: The MS ENG shall demonstrate a thorough understanding of the theory and application of Fresh Water Production Systems.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Definition and Purpose**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
2	**History and Development**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
3	**Theoretical Overview and Terminology**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
3.1	Desalination Principles: 1. Reverse osmosis;			

Teaching Points		Instructional Strategies	Time (min)	References
	2. Distillation; and 3. Advantages/disadvantages			
3.2	Treatment: 1. Chlorination; 2. Bromination; and 3. UV.			
3.3	Quality: 1. Feed; and 2. Potable			
4	**Components:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
4.1	Production: 1. Capacity; 2. Reverse osmosis: i. Membranes; ii. Pumps; iii. Filters; iv. Salinometers; and v. Dump valves. 3. Flash evaporators: i. Evaporator: 1. Demisters; 2. Ejectors; 3. Heaters; 4. Stages; and 5. Bursting disc. ii. Condenser; and iii. Pumps.			
5	**Starting and Stopping**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
6	**Running Parameters and Protection**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
7	**System Tests**	**Method:** Lecture		

Teaching Points		Instructional Strategies	Time (min)	References
		Media: PowerPoint **Environment:** Classroom		
7.1	Technician certification			
8	**Troubleshooting and Maintenance Considerations**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
8.1	Water quality			
8.2	Descaling			
9	**Corrosion and Preservation**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
10	**Safety and Environmental Considerations**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		

5. Total EO Time: Total: 6 x 50 minute periods as follows:

    a. 5 x 50 min period – Classroom; and

    b. 1 x 50 min periods – Workshop demonstration.

6. Training and Learning Aids:

    a. Static display of a Reverse Osmosis Desalination (ROD) plant.

7. Remarks: Nil.

PO 002 – MANAGE THE OPERATION OF AUXILIARY SYSTEMS

EO 002.08 DESCRIBE THE THEORY AND APPLICATION OF STEAM SYSTEMS

1. Performance: Describe the Theory and Application of Steam Systems.

2. Conditions:

   a. Given:

      (1) Supervision; and

      (2) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) N/A.

3. Standard: The MS ENG shall demonstrate a thorough understanding of the theory and application of Auxiliary and Domestic Steam Systems.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Definition and Purpose:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
2	**History and Development**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
3	**Advantages and Disadvantages:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		

Teaching Points		Instructional Strategies	Time (min)	References
3.1	Comparison with electric generation.			
4	**Components:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
4.1	Boilers 1. Coils; 2. Casing; 3. Furnace and brickwork; 4. Igniters; 5. Relief valves; 6. Stop valve; 7. Control system; 8. Air supply; 9. Fuel supply; 10. Feed water supply; and 11. Feed water tank;			
4.2	Steam Generator 1. Heating element; 2. Relief valves; and 3. Control system.			
4.3	Capacity considerations;			
5	**Flash-up and Shutdown**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
6	**Running Parameters and Protection:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
6.1	Parameters monitored;			
6.2	Warnings, alarms and trips;			
7	**Troubleshooting and Maintenance Considerations:**	**Method:** Lecture **Media:** PowerPoint		

Teaching Points		Instructional Strategies	Time (min)	References
		Environment: Classroom		
7.1	Coil replacement;			
7.2	Brickwork;			
7.3	Boiler injectors;			
7.4	Safety relief valves; and			
7.5	Water hammering;			
8	**Corrosion and Preservation:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
8.1	Boiler compound and feed water quality;			
9	**Safety and Environmental Considerations**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
10	**In-Service Systems:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
10.1	HFX Class;			
10.2	HDW Class; and			
10.3	PRO Class.			

5. Total EO Time: Total: 3 x 50 minute periods as follows:

   a. 2 x 50 min period – Classroom; and

   b. 1 x 50 min periods – EC (indicate total time or per student).

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 002 – MANAGE THE OPERATION OF AUXILIARY SYSTEMS

EO 002.09 DESCRIBE THE THEORY AND APPLICATION OF FUEL OIL FILLING AND TRANSFER SYSTEMS

1. Performance: Describe the Theory and Application of Fuel Oil Filling and Tranfer Systems.

2. Conditions:

   a. Given:

      (1) Supervision; and

      (2) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) N/A.

3. Standard: The MS ENG shall demonstrate a thorough understanding of the theory and application of Fuel Oil Filling and Transfer Systems.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Definition and Purpose**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
2	**Components**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
2.1	Pumps			
2.2	Tanks			
2.3	Applications: 1. Isolation; and 2. Control: i. Regulating; and ii. Relief			

Teaching Points		Instructional Strategies	Time (min)	References
	3. Packing			
2.4	Centrifuges			
2.5	Piping arrangement			
2.6	Valves			
2.7	Upper deck fittings: 1. Receiver bell; 2. NATO B; and 3. JP5			
2.8	Hoses			
2.9	Liquid Level Management System (LLMS)			
2.10	Water Displaced Fuel System (WDFS): 1. Overboard discharge; 2. Chain tanks; 3. Tank stripping arrangements; and 4. Oil content monitor			
3	**Troubleshooting and Maintenance Considerations**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
3.1	Fuel conditioning directive (polishing)			
3.2	Microbiological Growth (MBG)			
3.3	Tank stripping			
4	**Safety and Environmental Considerations**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
4.1	Maximum/Minimum Tank Capacity directives			
4.2	NAVORDs			
4.3	Environmental Management System			
6	**In-Service Systems (with emphasis on propellers):**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
6.1	HAL Class			

Teaching Points		Instructional Strategies	Time (min)	References
6.2	HDW Class			
6.3	PRO Class			

5. Total EO Time: Total: 2 x 50 minute periods as follows:

   a. 2 x 50 min period – Classroom.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 002 – MANAGE THE OPERATION OF AUXILIARY SYSTEMS

EO 002.10 DESCRIBE THE THEORY AND APPLICATION OF AIRCRAFT HANDLING AND FUEL SYSTEMS

1. Performance: Describe the Theory and Application of Aircraft Handling and Fuel Systems.

2. Conditions:

   a. Given:

      (1) Supervision; and

      (2) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) N/A.

3. Standard: The MS ENG shall demonstrate a thorough understanding of the theory and application of Aircraft Handling and Fuel Systems.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Definition and Purpose:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
1.1	Hauldown;			
1.2	Securing;			
1.3	Aligning; and			
1.4	Traversing.			
2	**History and Development:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
2.1	Bear trap; and			
2.2	Advantages and disadvantages.			

Teaching Points		Instructional Strategies	Time (min)	References
3	**Components:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
3.1	Aircraft handling: 1. Requirements: i. Recovery; ii. Securing; and iii. Traversing. 2. Flight deck; 3. Traps; 4. Power packs; 5. Accumulators; 6. Controls and instrumentation; i. LSO Console; ii. Local Operating console; iii. Flyco console; and iv. Components: 1. Motors; 2. Controllers; 3. Solenoids; 4. Load Cells; 5. Indication; 6. Moog valve; and 7. Strip chart recorder. v. Traversing system; vi. Dual winch and drum unit; vii. Reeving system: 1. Forward and aft traverse cables; 2. Traction cable; viii. Rapid securing device: 1. Frame; 2. Traverse shuttle;			

Teaching Points		Instructional Strategies	Time (min)	References
	3. Arresting beams; 4. Chain shuttles; 5. Brake system; and 6. Pneumatic system. ix. Tail guide winder cables. x. Brakes; xi. Securing device; xii. Cables; xiii. Hangar; xiv. Flight deck and signal lighting; and xv. External interfaces: 1. Electrical power requirements; 2. HP/LP air requirements; 3. Mounting arrangements; and 4. Operator requirements.			
3.2	Fueling: 1. Requirements: i. Fueling: 1. Quality/ recirculation; 2. Capacity; 3. HIFR; 4. Hot/Cold; 5. JP5/JP4; and 6. Grounding. ii. Pumps; iii. Tanks;			

Teaching Points		Instructional Strategies	Time (min)	References
	iv. Heaters; v. Supply/return lines; vi. Filters/coalescers; vii. Flow meters; viii. HIFR hoses; and ix. Flight deck track fuel dams.			
4	**Operational Considerations:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
4.1	Shipboard Helicopter Operating Procedures (SHOPs)			
5	**Running Parameters and Protection:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
5.1	Electrical load;			
5.2	Pneumatic pressures;			
5.3	Oil pressure;			
5.4	Maximum capacity/load; and			
5.5	Other relevant parameters.			
6	**Running Parameters and Protection:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
6.1	Parameters monitored;			
6.2	Warnings, alarms and trips;			
7	**Troubleshooting and Maintenance Considerations:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
7.1	Coil replacement;			
7.2	Brickwork;			
7.3	Boiler injectors;			
7.4	Safety relief valves; and			
7.5	Water hammering;			
8	**System Tests and Trials:**	**Method:** Lecture		

Teaching Points		Instructional Strategies	Time (min)	References
		Media: PowerPoint **Environment:** Classroom		
8.1	Flight Deck Certification.			
9	**Troubleshooting Maintenance Considerations:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
9.1	Recirculation;			
9.2	Pre-fueling test procedures; and			
9.3	DAND samples.			
10	**Corrosion and Preservation**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
11	**In-Service Systems:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
11.1	HAL Class			
11.2	HDW Class			
11.3	PRO Class			

5. Total EO Time: Total: 10 x 50 minute periods as follows:

   a. 7 x 50 min period – Classroom; and

   b. 3 x 50 min periods – Lab Practical.

6. Training and Learning Aids:

   a. Operational aircraft handling system.

7. Remarks: Nil.

PO 002 – MANAGE THE OPERATION OF AUXILIARY SYSTEMS

EO 002.11 DESCRIBE THE THEORY AND APPLICATION OF WASTE MANAGEMENT SYSTEMS

1. Performance: Describe the Theory and Application of Waste Management Systems.

2. Conditions:

   a. Given:

      (1) Supervision; and

      (2) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) N/A.

3. Standard: The MS ENG shall demonstrate a thorough understanding of the theory and application of Waste Management Systems.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Definition and Purpose**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
2	**History and Development**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
3	**Components**	**Method:** Lecture **Media:** PowerPoint		

Teaching Points		Instructional Strategies	Time (min)	References
		Environment: Classroom		
3.1	Solid waste handling systems: 1. Plastics; and 2. Organics: 3. Paper; and 4. Solid food waste. 5. Metal; 6. Glass; 7. Collection/separation; and 8. Storage/disposal.			
3.2	Bilge water processing systems: 1. Pumps; 2. Tanks; 3. Filtration; 4. Separator; 5. Oil content monitor; 6. Overboard discharge; and 7. Upper deck fittings			
4	**Troubleshooting and Maintenance Considerations**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
5	**Safety and Environmental Considerations**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
5.1	Overboard discharge direction			
5.2	NAVORDs			
5.3	Environmental Management System			
6	**Administration**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
6.1	Permission to discharge			
6.2	Logs			

Teaching Points		Instructional Strategies	Time (min)	References
7	**In-Service waste management systems**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
7.1	Solid Waste system			
7.2	Sarex			
7.3	Hydromen			

5. Total EO Time: Total: 4 x 50 minute periods as follows:

   a. 2 x 50 min period – Classroom;

   b. 1 x 50 min period – Workshop demonstration; and

   c. 1 x 50 min periods – EC (indicate total time or per student).

6. Training and Learning Aids:

   a. Static display of Bilge water processing system.

7. Remarks: Nil.

PO 002 – MANAGE THE OPERATION OF AUXILIARY SYSTEMS

EO 002.12 DESCRIBE AUXILIARY SYSTEMS IN HMC SHIPS

1. Performance: Describe Auxiliary Systems in HMC Ships.

2. Conditions:

   a. Given:

      (1) Operational Ship;

      (2) Supervision; and

      (3) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) N/A.

3. Standard: The MS ENG shall demonstrate a thorough understanding of the theory and application of Waste Management Systems.

4. Teaching Points:

The MS ENG shall produce diagrams and written reports in order to explain the functions of the following equipment and systems as fitted on board for the applicable class of ship:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Refrigeration systems:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
1.1	Main refrigeration plant; and			
1.2	Chilled water system.			
2	**Compressed air systems:**	**Method:** Lecture **Media:** PowerPoint		

Teaching Points		Instructional Strategies	Time (min)	References
		Environment: Classroom		
2.1	HP air system;			
2.2	LP air system; and			
2.3	Breathable air systems.			
3	**Steering system:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
3.1	Control system; and			
3.2	Actuating system.			
4	**Auxiliary sea water system**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
5	**Fresh water production systems**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
6	**Steam systems:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
6.1	Auxiliary steam system; and			
6.2	Domestic steam system: 1. Boilers; 2. Steam generators; 3. Distribution; and 4. Returns.			
7	**Fuel oil systems:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
7.1	Ship's tank layout;			
7.2	Filling systems;			

Teaching Points		Instructional Strategies	Time (min)	References
7.3	Transfer systems; and			
7.4	RAS equipment and layout.			
8	**Aircraft handling/fuel systems**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
8.1	JP5 Fuel System; and			
8.2	RAST.			
9	**Waste management systems:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
9.1	Solid Waste; and			
9.2	Bilge water processing.			

5. Total EO Time: Throughout the Afloat phase.

6. Training and Learning Aids: Nil.

7. Remarks: The MS ENG shall produce drawing that are:

   a. Neat/legible;

   b. Consistent with ship's fitted drawings; and

   c. Complete, to include as applicable:

      (1) Content and capacities;

      (2) Direction of flow;

      (3) Monitoring arrangements and parameters;

      (4) Control arrangements;

      (5) Valve types; and

      (6) A uniform colour-code.

d. Consistent with CFTO C-23-007-001/MD-001, Ship's Piping Systems as the standard for symbols.

PO 002 – MANAGE THE OPERATION OF AUXILIARY SYSTEMS

EO 002.13 DESCRIBE THE OPERATION OF AUXILIARY SYSTEMS IN HMC SHIPS

1. Performance: Describe the Operation of Auxiliary Systmes in HMC Ships.

2. Conditions:

   a. Given:

      (1) Operational Ship;

      (2) Supervision; and

      (3) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) All conditions, day and night.

3. Standard: The MS ENG shall demonstrate a thorough understanding of the operation of the Auxiliary Systems fitted on board.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**The MS ENG shall participate in all modes of operation (including flash up and shutdown) of auxiliary equipment and systems fitted on board to include:**	**Method:** OJT **Media:** N/A **Environment:** Ship		
1.1	Refrigeration systems: 1. Main refrigeration plant; and 2. Chilled water system			
1.2	Compressed air systems: 1. HP air system; 2. LP air system; and			

Teaching Points		Instructional Strategies	Time (min)	References
	3. Breathable air systems			
1.3	Steering system: 1. Control system; and 2. Actuating system			
1.4	Auxiliary sea water system			
1.5	Fresh water production system			
1.6	Steam systems: 1. Auxiliary steam system; and 2. Domestic steam system: 3. Boilers; 4. Steam generator; 5. Distribution; and 6. Returns			
1.7	Fuel oil systems: 1. Ship's tank layout; 2. Filling systems; 3. Transfer systems; and RAS equipment and layout			
1.8	Aircraft handling/fuel systems: 1. JP5 Fuel System; and 2. RAST			
1.9	Waste management systems: 1. Solid waste management; and 2. Bilge water processing			
2	**The MS ENG shall produce written reports on the operation of the auxiliary equipment and systems listed in para 10a to include:**	**Method:** OJT **Media:** N/A **Environment:** Ship		
2.1	Normal operation			
2.2	Abnormal operations			
2.3	Emergency operations			
2.4	Operating parameters			
2.5	Operational limitations			
2.6	Monitoring methods			

Teaching Points		Instructional Strategies	Time (min)	References
2.7	Flash ups			
2.8	Shutdowns			
2.9	Safety considerations			
2.10	Troubleshooting			
2.11	Routine running maintenance			

5. Total EO Time: Throughout the Afloat phase.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 003 – MANAGE THE OPERATION OF CONTROL SYSTEMS

EO 003.01 DESCRIBE THE THEORY AND APPLICATION OF CONTROL SYSTEMS

1. Performance: Describe the Theory and Application of Control Systems.

2. Conditions:

   a. Given:

      (1) Supervision; and

      (2) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) N/A.

3. Standard: The MS ENG shall demonstrate a thorough understanding of the theory and application of Control Systems.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Definition and Purpose:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
2	**History and Development**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
3	**Advantages and Disadvantages:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		

Teaching Points		Instructional Strategies	Time (min)	References
3.1	Advantages: 1. Reduced manpower; 2. Reduced human error; 3. Faster and standardized response; 4. Continuous monitoring of systems; and 5. Decreased life cycle cost.			
3.2	Disadvantages: 1. Specialized maintainers (higher cost); 2. Complacency of watchkeepers; and 3. Loss of knowledge of machinery and systems by operators.			
4	**Theoretical Overview and Terminology:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
4.1	Dynamic characteristics;			
4.2	System specifications;			
4.3	The control loop: 1. Open Loop: i. Components: 1. Final element (actuator); and 2. Process. ii. Internal signals: 1. Manipulated variable; and 2. Controlled variable. iii. External interfaces: 1. Power sources; and 2. Set point (input or desired value). 2. Closed Loop (automatic control system): i. Components: 1. Comparator;			

Teaching Points		Instructional Strategies	Time (min)	References
	2. Controlling element; 3. Final element (actuator); 4. Process; 5. Primary element (sensor, transducer); 6. Measuring element; and 7. Indication element (display, gauge). ii. Signals: 1. Error; 2. Actuating; 3. Manipulated variable; 4. Controlled variable; 5. Measured variable; and 6. Indicated variable. iii. External interfaces: 1. Set point; 2. Power sources; and 3. Disturbances to the process.			
4.4	Modes of operation: 1. Modes of operation of a generic control system: i. Automatic control ii. Servo manual control or remote control iii. Manual control			
4.5	Frequency response and system compensation;			
4.6	Synchros/servos;			
4.7	Transducers; and			
4.8	Input types (set points):			

Teaching Points		Instructional Strategies	Time (min)	References
	1. Constant or regular inputs such as refrigerated storage room temperature, and 2. Variable or follow-up inputs which require the controlled variable to follow a frequently changing reference input such as ships speed demands from command. To include such types of input as: i. Step (the steady state is a constant input); ii. Ramp; iii. Trigonometric (i.e. sinusoidal); iv. Exponential; v. Pulsed; and vi. Irregular.			
4.9	Types: 1. Pneumatic; 2. Hydraulic; 3. Mechanical; 4. Electrical/Electronic: i. analog; and ii. digital. 5. Optical; and 6. Combinations of above.			
4.10	Types of Response: 1. Corrective action - senses changes and corrects, for example: propeller pitch or boiler water level; 2. Alarm action - senses changes and alerts the operator, for example: lube oil pressure; 3. Shut down action - senses changes and shuts down the system, for example: fridges; and			

Teaching Points		Instructional Strategies	Time (min)	References
	4. Combined action - senses changes and combines the actions, for example: low lube oil pressure sounding an alarm and shutting down a diesel engine.			
4.11	Controller action: 1. Control action: i. On-off action, control (position) of the final element is limited to two positions: 1. On/off; or 2. Fully open/fully closed. ii. Proportional action, control of the final element varies in proportion to the magnitude of the error signal. Discussion to include but not limited to the following terms: 1. Proportional band/gain; 2. Offset; 3. Set point; and 4. Sensitivity. iii. Integral action, control of the final element varies in relation to the length of time that the error has been present. Discussion to include but not limited to the following:			

Teaching Points		Instructional Strategies	Time (min)	References
	1. Reset/reduction of offset; 2. Stability; 3. Proportional plus integral action; and 4. Integral action time. iv. Derivative action, control of the final element varies in relation to rate of change of the error. Discussion to include but not limited to the following: 1. Initial action/overcoming system inertia; 2. Stability; and 3. Proportional plus derivative action. v. Combined action, control of the final element varies in relation to a combination of the magnitude, rate of change and the time of the error. Discussion to include but not limited to the following: 1. Adjustment/tuning of: a. Set point; and b. Gain/proportional band. 2. Response time:			

Teaching Points		Instructional Strategies	Time (min)	References
	a. Transient; and b. Steady state. 3. Stability/damping: a. No damping; b. Underdamped; c. Critically damped; and d. Overdamped. 4. Time constant; 5. Oscillations; 6. Backlash; and 7. Phase lag/lead.			
4.12	Equipment requirement;			
4.13	Control hierarchy;			
4.14	Actuating medium;			
4.15	Speed/precision;			
4.16	Graceful degradation;			
4.17	Ergonomics (MMIs);			
4.18	Computer technology including principles of data transfer, data communication, and host-target systems relationships;			
5	**Components**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
5.1	Human-Machine Interface (HMI);			
5.2	Processor;			
5.3	Receiver/transmitter;			

Teaching Points		Instructional Strategies	Time (min)	References
5.4	Feedback;			
5.5	Instrumentation;			
5.6	Sensors;			
5.7	Controlled element;			
5.8	Actuator;			
5.9	Storage device;			
5.10	Software;			
5.11	I/O devices; and			
5.12	Power supplies.			
6	**Troubleshooting and Maintenance Considerations:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
6.1	Software upgrades; and			
6.2	Password control.			
7	**In-Service Machinery Control & Platform Management Systems:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
7.1	Integrated Platform Management System (IPMS): 1. Design objectives: i. Objectives and requirements in the design of IPMS to include: 1. Survivability; 2. Reliability; 3. Modularity (maintainability and operational flexibility); 4. Standardization; 5. Graceful degradation (vital functions maintained); 6. Flexibility;			

Teaching Points		Instructional Strategies	Time (min)	References
	7. Redundancy; 8. Reduced manning; 9. Greater capability; 10. Improvability; 11. Reduced maintenance costs; 12. Distributed architecture; and 13. Onboard training capability. 2. Equipment monitored and controlled by control system; i. Propulsion machinery: 1. Gas turbine engines; 2. Propulsion diesel engine; and 3. Gearbox and shafting. ii. Ancillary equipment; iii. Auxiliary machinery; an iv. Electrical plant: 1. Primary circuit breakers; 2. Prime movers and electrical generators; 3. Switchboards; 4. Power generation and distribution; and 5. Electrical miscellaneous. 3. Human-Machinery Interfaces (HMIs);			

Teaching Points		Instructional Strategies	Time (min)	References
	i. Bridge control console (BCC): 1. Automatic propulsion control; 2. Telegraph orders; 3. Monitoring of propulsion machinery; 4. Visual and audible alarm, warning and device error information. ii. Machinery control console (MCC): 1. Automatic propulsion control; 2. Semi-automatic propulsion control; 3. Control and monitoring of propulsion, ancillary, auxiliary and electrical machinery; 4. Visual and audible alarm, warning and device error information; and 5. Control of the video hardcopier. iii. Supervisor's console (SC): 1. Display of machinery health and performance monitoring data/adjustment of health			

Teaching Points		Instructional Strategies	Time (min)	References
	monitoring parameters; 2. Automatic propulsion control; 3. Semi-automatic propulsion control; 4. Control and monitoring of propulsion, ancillary and auxiliary machinery; 5. Visual and audible alarm, warning and device error information; and 6. Control of the video hardcopier. iv. Maintainer's/electrical console (MEC): 1. IPMS maintenance and diagnostic functions; 2. Monitoring of propulsion machinery; 3. Control and/or monitoring of ancillary, auxiliary and electrical machinery; 4. Visual and audible alarm, warning and device error information; and 5. Control of the video hardcopier.			

Teaching Points		Instructional Strategies	Time (min)	References
	v. Large Screen Displays (LSD); vi. Portable Operating Units (POUs); vii. Local operating panels including: 1. Automatic propulsion control; 2. Manual propulsion control; 3. Control and monitoring of propulsion and vital ancillary machinery; and 4. Visual and audible alarm, warning and device error information. viii. Group peripheral: 1. Equipment/ printer station. ix. Machinery controllers (MC): 1. Propulsion management; 2. Sequence control of ancillary and auxiliary; and 3. Coordinating communication between sub-systems over the fiber optic network. 4. Remote terminal units (RTUs); i. Plant electrical interface; ii. Transducer excitation; iii. Signal conditioning and processing;			

Teaching Points		Instructional Strategies	Time (min)	References
	iv. Isolation; v. Power conditioning, distribution and conversion; vi. Signal limit tests; vii. IPMS interface; viii. Plant control; and ix. Diagnostics. 5. Engine Control Modules (ECMs); i. Start and stop sequencing; ii. Engine protection; iii. Control and monitoring of ancillary equipment; iv. Vibration monitoring; and v. Quick response local controls. 6. Fiber-Optic Network; 7. Serial links and Network Switches; 8. Equipment Health monitoring (HM) unit; i. Periodic logging; ii. Running hours; iii. Vibration analysis; iv. Gas path analysis; v. Trend analysis; and vi. Dynamic analysis. 9. On-Board Training System (OBTS); and 10. Peripheral devices; 11. Field devices (actuation and feedback), to include: i. Power Lever Angle (PLA); and ii. CRPP. 12. Software/Firmware.			
7.2	HAL Class – Battle Damage Control System (BDCS):			

Teaching Points		Instructional Strategies	Time (min)	References
	1. Damage Control Console (DCC) layout; i. HQ1; ii. HQ2; iii. Bridge alarm panel; iv. QM alarm panel; and v. Ops Room alarm panel. 2. Bus Access Modules (BAMs); 3. Field devices; and 4. Panel components (conditions monitored by each): i. Fuel filling and transfer/liquid level management system; ii. Fire detection and suppression alarm control system; iii. Heating, ventilation and air conditioning (HVAC) systems; and iv. Firemain and pumps system. 5. IPMS - DCS Interface. i. Connection points between IPMS and legacy DCS System BAM Loops.			
7.3	HDW Class – Battle Damage Control System (BDCS): **[To be completed].**			
7.4	PRO Class – Battle Damage Control System (BDCS): **[To be completed].**			

5. Total EO Time: Total: 58 x 50 minute periods as follows:

   a. 35 x 50 min period – Classroom;

   b. 7 x 50 min period – Lab practical;

   c. 7 x 50 min period – IPMS practical;

   d. 7 x 50 min period – BDCS practical; and

   e. 2 x 50 min periods – EC (indicate total time or per student).

6. Training and Learning Aids:

   a. Control systems lab, IPMS test cell and BDCS test cell.

7. Remarks:

   a. Control systems lab practical to reinforce control systems principles instructed. IPMS and BDCS test cell to instruct operation of both systems.

PO 003 – MANAGE THE OPERATION OF CONTROL SYSTEMS

EO 003.02 DESCRIBE CONTROL SYSTEMS IN HMC SHIPS

1. Performance: Describe Control Systems in HMC Ships.

2. Conditions:

   a. Given:

      (1) Operational Ship;

      (2) Supervision; and

      (3) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) All conditions, day and night.

3. Standard: The MS ENG shall demonstrate a thorough understanding of the Machinery Control & Platform Management Systems fitted on board.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**The MS ENG shall produce detailed diagrams and written reports in order to explain the functions of the following equipment and systems as fitted on board for the applicable class of ship:**	**Method:** OJT **Media:** N/A **Environment:** Ship		
1.1	Machinery Control & Platform Management Systems (IPMS): Human Machinery Interfaces (HMIs);			

Teaching Points		Instructional Strategies	Time (min)	References
	Remote Terminal Units (RTUs); Fiber Optic Networks; Local Operating Panels (LOPs); Serial Links and Network Switches; Equipment Health Monitoring (HM) Unit; On-Board Training System (OBTS); Peripheral Devices; and Field Devices (actuation and feedback).			
1.2	Propulsion Control Algorithm (narrative only)			
1.3	Battle Damage Control System (BDCS)			

5. Total EO Time: Throughout the Afloat phase.

6. Training and Learning Aids: Nil.

7. Remarks: The MS ENG shall produce drawings that are:

   a. Neat/legible;

   b. Consistent with ship's fitted drawings; and

   c. Complete, to include as applicable:

      (1) System ratings;

      (2) Data/Information pathways;

      (3) Monitoring arrangements and parameters;

(4) Control arrangements; and

(5) A uniform colour-code.

PO 003 – MANAGE THE OPERATION OF CONTROL SYSTEMS

EO 003.03 DESCRIBE THE OPERATION OF CONTROL SYSTEMS IN HMC SHIPS

1. Performance: Describe the Operation of Control Systems in HMC Ships.

2. Conditions:

   a. Given:

      (1) Operational Ship;

      (2) Supervision; and

      (3) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) All conditions, day and night.

3. Standard: The MS ENG shall demonstrate a thorough understanding of the operation of Control Systems fitted on board.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**The MS ENG shall participate in all modes of operation of Control Systems fitted on board to include:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
1.1	Machinery Control System; and			
1.2	Damage Control System.			
2	**The MS ENG shall produce written reports on the operation of the control systems listed in para 10a to include:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		

Teaching Points		Instructional Strategies	Time (min)	References
2.1	Normal operation;			
2.2	Abnormal operations;			
2.3	Monitoring methods;			
2.4	Safety considerations; and			
2.5	Trouble shooting.			

5. Total EO Time: Throughout the Afloat training phase.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 004 – MANAGE THE OPERATION OF ELECTRICAL SYSTEMS

EO 004.01 DESCRIBE THE THEORY AND APPLICATION OF ELECTRICAL SYSTEMS

1. Performance: Describe the Theory and Application of Electrical Systems.

2. Conditions:

   a. Given:

      (1) Supervision; and

      (2) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) N/A.

3. Standard: The MS ENG shall demonstrate a thorough understanding of the theory and application of Electrical Systems.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Definition and Purpose**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
2	**Basic Electrical Theory**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
2.1	DC circuits			
2.2	AC circuits			
2.3	Semi-conductors			
2.4	Circuit analysis			
2.5	DC generators			
2.6	AC generators			

Teaching Points		Instructional Strategies	Time (min)	References
2.7	MG sets			
2.8	Static convertors			
2.9	Induction motors			
2.10	Synchronous motors			
2.11	DC motors			
2.12	Controllers			
2.13	Protection devices			
2.14	Transformers			
2.15	Batteries			
2.16	Electrical instrumentation			
2.17	Switching			
3	**Cathodic Protection**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
3.1	Underwater corrosion process			
3.2	Impressed current/sacrificial			
3.3	Controller			
3.4	Reference electrodes			
3.5	Anodes			
3.6	Rectifier			
4	**Degaussing:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
4.1	Cause of ship's magnetization: 1. Earth's magnetic field; 2. Influence of Earth's magnetic field during construction; and 3. Influence of Earth's magnetic field during operation			
4.2	Hazards due to ship magnetization: 1. Principle of operation of magnetic mines; and			

Teaching Points		Instructional Strategies	Time (min)	References
	2. Ship's magnetic signature before degaussing			
4.3	Purpose of 'M' coil: 1. Ship's signature without 'M' coil; and 2. Ship's signature with 'M' coil			
4.4	Reason for 3 part 'M' coil: 1. 'M' coil misfit; and 2. Ship's signature with 3 part 'M' coil			
4.5	Purpose of 'FI' and 'QI' coils: 1. Longitudinal magnetization; 2. Ideal solution; and 3. How 'FI' and 'QI' coils affect longitudinal magnetization			
4.6	Purpose of 'AI' coils and why these coils are only used on large ships			
4.7	General location of ship's degaussing coils			
4.8	Description of ship's degaussing generators: 1. Construction; and 2. Characteristics			
4.9	Description of degaussing switchboard: Controls; and Configuration			
4.10	Application of required current for coils 1. Determination of correct current value for each coil; and			

Teaching Points		Instructional Strategies	Time (min)	References
	2. Procedure for applying correct settings on control panel			
4.11	Purpose of gyro input: 1. Effect of changing course on ILM; 2. 3 step course correction in automatic mode; and 3. 3 step course correction in manual mode			
4.12	Objective of the degaussing trial			
4.13	Conducting authority for the trial;			
4.14	Facilities available: 1. Bedford Basin, Halifax; 2. McNab's Island, Halifax; and 3. Esquimalt			
4.15	General Description of the trial procedure: 1. Preparations for trial; 2. Trail procedures; and 3. Means for requesting trial.			
5	**Deperming**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
5.1	Theory			
5.2	Preparations: 1. Coil wraps; 2. Smart card removals; and 3. Reduced manning			
6	**Shaft grounding**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
6.1	Passive; and			

Teaching Points		Instructional Strategies	Time (min)	References
	1. Active			
7	**Communication systems (of the MSE Department)**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
7.1	Sound power			
8	**Helo start and service system**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
8.1	Motor generators			
8.2	SFCs			
8.3	Cables			
8.4	Receptacle			
9	**Electrical Safety**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
9.1	Personal protective equipment (PPE)			
9.2	Safe working practices			
9.3	Tagging out procedures			

5. Total EO Time: Total: 40 x 50 minute periods as follows:

   a. 28 x 50 min period – Classroom; and

   b. 12 x 50 min period – Lab practical;

6. Training and Learning Aids:

   a. Electrical lab.

7. Remarks: Nil.

PO 004 – MANAGE THE OPERATION OF ELECTRICAL SYSTEMS

EO 004.02 DESCRIBE THE THEORY AND APPLICATION OF POWER GENERATION AND DISTRIBUTION (PG&D) SYSTEMS

1. Performance: Describe the Theory and Application of Power Generation and Distribution (PG&D) Systems.

2. Conditions:

   a. Given:

      (1) Supervision; and

      (2) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) N/A.

3. Standard: The MS ENG shall demonstrate a thorough understanding of the theory and application of Power Generation and Distribution Systems.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Definition and Purpose**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
2	**Theoretical Overview and Terminology:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
2.1	Principles of 3 phase power generation;			
2.2	Generator characteristics;			
2.3	Paralleling;			

Teaching Points		Instructional Strategies	Time (min)	References
2.4	Voltage regulation; and			
2.5	Governors.			
2.6	Load requirements;			
2.7	Distribution;			
2.8	Switching;			
2.9	Vital and non-vital loads;			
2.10	Earth detection; and			
2.11	Power factor.			
3	**Components:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
3.1	Main switchboards;			
3.2	Emergency switchboards;			
3.3	Generator set;			
3.4	Voltage regulators;			
3.5	Governors;			
3.6	Transformers; and			
3.7	Frequency converters: 1. SFCs; and 2. Motor generators.			
3.8	Transfer switches (ACOS, MCOS, ABT);			
3.9	Shore power connections;			
3.10	Bus ties;			
3.11	Casualty power connections;			
3.12	Circuit identification; and			
3.13	Power distribution: 1. Primary: a. NATO standard. 2. Secondary: a. 110v, 3phase, 60hz; b. 450v, 3 phase, 400hz;			

Teaching Points		Instructional Strategies	Time (min)	References
	c. 110v, 3 phase, 400hz; and d. 28v DC.			
4	**Operational Considerations:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
4.1	Load shedding;			
4.2	Power failure;			
4.3	Emergency power isolation; and			
4.4	Stable power.			
5	**Starting and stopping:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
5.1	Pre-start checks;			
5.2	Start permissives;			
5.3	Auto start sequence;			
5.4	Starts: 4. Manual; and 5. Emergency.			
5.5	Stopping: 2. Normal; and 3. Emergency.			
6	**Running Parameters and Protection:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
6.1	Parameters monitored;			
6.2	Warnings, alarms and trips; and			
6.3	Fire detection and suppression.			

Teaching Points		Instructional Strategies	Time (min)	References
7	**Trials:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
7.1	Load bank.			
8	**Installation and Removal Considerations:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
8.1	Generator replacement.			
9	**Troubleshooting and Maintenance Considerations:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
9.1	Insulation checks.			
10	**Corrosion and Preservation**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
11	**Safety and Environmental Considerations**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
12	**In-Service systems:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
12.1	HAL Class;			
12.2	HDW Class;			
12.3	PRO Class;			
12.4	KIN Class; and			
12.5	VIC Class.			

5. Total EO Time: Total: 35 x 50 minute periods as follows:

   a. 28 x 50 min period – Classroom; and

   b. 7 x 50 min period – Workshop demonstration;

6. Training and Learning Aids:

   a. Static displays of typical generation electrical equipment.

7. Remarks: Nil.

PO 004 – MANAGE THE OPERATION OF ELECTRICAL SYSTEMS

EO 004.03 DESCRIBE SHIP'S SIGNATURE

1. Performance: Describe Ship's Signature.

2. Conditions:

   a. Given:

      (1) Supervision; and

      (2) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) N/A.

3. Standard: The MS ENG shall demonstrate a thorough understanding of ship's signature.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Signature principles**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
1.1	Theory			
1.2	Definition and purpose			
1.3	Characteristics			
1.4	Normal/Abnormal signature limits			
2	**Signature types**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
2.1	Acoustic			
2.2	Magnetic			

Teaching Points		Instructional Strategies	Time (min)	References
2.3	Infrared			
2.4	Radar Cross Section (RCS)			
3	**Policy and procedures**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
3.1	NAVORD 3395-1 Naval Signature Management Policy and Organization			
3.2	CFCD 109 (vol. 1-4)			
4	**Organizations**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
4.1	Ship			
4.2	FER			
4.3	FMF			
4.4	MSHQ			
5	**Types of ranging**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
5.1	In-water acoustic			
5.2	In-water non-acoustic			
5.3	Infrared			
6	**Noise generation**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
6.1	Machinery			
6.2	Hull			
6.3	Mounts			
6.4	Crew			
6.5	Shaft line/propeller			
7	**Reduction methods**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		

Teaching Points		Instructional Strategies	Time (min)	References
7.1	Ship's staff			
7.2	Active shaft grounding			
7.3	Degaussing			
7.4	Deperming			
7.5	Radar Absorbent Paint			
7.6	RAM Pads			
7.7	Anecoic Tile			
7.8	Ship's aspect			
8	**Quiet running states**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
8.1	Normal Quiet			
8.2	Patrol Quiet			
8.3	Ultra Quiet			
9	**Quiet running bill**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
9.1	Machinery state			
9.2	Habitability considerations			
10	**Maintenance requirements**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
10.1	Noise rounds			
10.2	Noise short recognition			
10.3	Noise short rectification			
10.4	Fitted Degaussing equipment.			

5. Total EO Time: Total: 4 x 50 minute periods as follows:

   a. 2 x 50 min period – Classroom; and

   b. 2 x 50 min periods – EC (indicate total time or per student).

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 004 – MANAGE THE OPERATION OF ELECTRICAL SYSTEMS

EO 004.04 DESCRIBE ELECTRICAL SYSTEMS IN HMC SHIPS

1. Performance: Describe Electrical Systems in HMC Ships.

2. Conditions:

   a. Given:

      (1) Operational Ship;

      (2) Supervision; and

      (3) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) All conditions, day and night.

3. Standard: The MS ENG shall demonstrate a thorough understanding of the Electrical Systems fitted on board.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Shipboard electrical power generation and distribution systems:**	**Method:** OJT **Media:** N/A **Environment:** Ship		
1.1	Generators;			
1.2	Prime movers; and			
1.3	Power generation and distribution system, to include: 1. Main switchboards; 2. Emergency switchboards; 3. Generators; 4. Transfer switches (ACOS, MCOS, ABT); 5. Shore power connections; 6. Bus ties; and			

Teaching Points		Instructional Strategies	Time (min)	References
	7. Casualty power connections.			
1.4	Secondary power distribution (narrative only): 1. 110v, 3 phase, 60hz; 2. 450v, 3 phase, 400hz (helo start and service system); 3. 110v, 3 phase, 400hz; and 4. 28 v DC.			
2	**Cathodic protection (narrative only)**	**Method:** OJT **Media:** N/A **Environment:** Ship		
3	**Degaussing/Deperming (narrative only)**	**Method:** OJT **Media:** N/A **Environment:** Ship		
4	**Communication systems (of the MSE Dept., narrative only)**	**Method:** OJT **Media:** N/A **Environment:** Ship		
5	**Shaft grounding (narrative only)**	**Method:** OJT **Media:** N/A **Environment:** Ship		
6	**Shipboard lighting (narrative only)**	**Method:** OJT **Media:** N/A **Environment:** Ship		
6.1	Navigation;			
6.2	Emergency; and			
6.3	Flight deck.			

5. Total EO Time: Throughout the Afloat training phase.

6. Training and Learning Aids: Nil.

7. Remarks: The MS ENG shall produce drawings that are:

   a. Neat/legible;

b. Consistent with ship’s fitted drawings; and

c. Complete, to include as applicable:

(1) System ratings;

(2) Electrical path;

(3) Monitoring arrangements and parameters

(4) Control arrangements; and

(5) A uniform colour-code.

PO 004 – MANAGE THE OPERATION OF ELECTRICAL SYSTEMS

EO 004.05 DESCRIBE THE OPERATION OF ELECTRICAL SYSTEMS IN HMC SHIPS

1. Performance: Describe the Operation of Electrical Systems in HMC Ships.

2. Conditions:

   a. Given:

      (1) Operational Ship;

      (2) Supervision; and

      (3) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) All conditions, day and night.

3. Standard: The MS ENG shall demonstrate a thorough understanding of the operation of the Electrical Systems fitted on board.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**The MS ENG shall participate in all modes of operation (including flash up and shutdown) of electrical equipment and systems fitted on board to include**	**Method:** OJT **Media:** N/A **Environment:** Ship		
1.1	Shipboard electrical power generation and distribution systems			
1.2	Cathodic protection			
1.3	Degaussing/deperming;			
1.4	Communications systems;			

Teaching Points		Instructional Strategies	Time (min)	References
1.5	Shaft grounding			
1.6	Shipboard lighting			
1.7	Helo start and service system			
2	**The MS ENG shall produce written reports on the operation of the electrical equipment and systems listed in para 10a to include (where applicable):**	**Method:** OJT **Media:** N/A **Environment:** Ship		
2.1	Normal operation			
2.2	Abnormal operations			
2.3	Emergency operations			
2.4	Operating parameters			
2.5	Operational limitations			
2.6	Monitoring methods			
2.7	Flash ups			
2.8	Shutdowns			
2.9	Safety considerations			
2.10	Troubleshooting			
2.11	Routine running maintenance			

5. Total EO Time: Throughout the Afloat training phase.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 005 – MANAGE THE OPERATION OF HULL SYSTEMS

EO 005.01 DESCRIBE THE THEORY AND APPLICATION OF HULL STRUCTURAL DESIGN, CONTRUCTION AND PRESERVATION

1. Performance: Describe the Theory and Application of Hull Systems Structures.

2. Conditions:

   a. Given:

      (1) Supervision; and

      (2) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) N/A.

3. Standard: The MS ENG shall demonstrate a thorough understanding of the theory and application of Hull Systems and Structural Design.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Definition and Purpose**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
2	**History and Development**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
3	**Theoretical Overview and Terminology:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
3.1	Forces acting on structure;			
3.2	Longitudinal strength considerations;			
3.3	Structure classification;			
3.4	Ship hydrodynamics:			

Teaching Points		Instructional Strategies	Time (min)	References
	1. Hull form: i. Hull form: ii. Short and wide; and iii. Single or multiple. 2. Resistance to movement through water: i. Hull condition; ii. Desired speed; iii. Trim; and iv. Displacement. 3. Interaction with propeller; and 4. Power vs speed curves.			
3.5	Classification societies;			
3.6	Role design considerations;			
3.7	Mission statement; and			
3.8	Design spiral: 1. Purpose; and 2. Criteria.			
3.9	Shipboard structural drawings 1. General arrangement.			
4	**Components:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
4.1	Keel/stem and stern/bottom;			
4.2	Side and shell plating;			
4.3	Longitudinals/beams/girders;			
4.4	Framing/inner/bottoms/weather decks/intermediate decks;			
4.5	Main transverse and longitudinal bulkheads;			
4.6	Superstructure/masts;			
4.7	Bilge keels/side docking keels;			
4.8	Pillars/supports;			
4.9	Sea tubes/inlet boxes;			
4.10	Beam/frame connections;			
4.11	Expansion joints;			

Teaching Points		Instructional Strategies	Time (min)	References
4.12	Ship structure interdependencies;			
4.13	Function of structural members; and			
4.14	Holes in structure.			
5	**Ship construction materials:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
5.1	Ferrous;			
5.2	Non-ferrous;			
5.3	Special alloys; and			
5.4	Other.			
6	**Construction methods:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
6.1	Welding: 1. Soldering and brazing; 2. Shielded metal arc; 3. TIG; and 4. MIG.			
6.2	Cutting: 1. Oxyacetylene; 2. Exothermic; and 3. Plasma.			
6.3	Welding considerations: 1. Procedures; and 2. Certification.			
7	**Painting:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
7.1	Why do we paint?			
7.2	Constituents;			
7.3	Types of paint;			
7.4	Epoxides;			
7.5	Anti-fouling;			
7.6	Surface preparation;			
7.7	Principles; and			

Teaching Points		Instructional Strategies	Time (min)	References
7.8	Methods.			
8	**Habitability:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
8.1	Thermal and acoustic insulation;			
8.2	Health and safety precautions;			
8.3	Material selection;			
8.4	Deck coverings; and			
8.5	Improvements.			
9	**Anchors and cables**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
10	**Test and Trials:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
10.1	Hull surveys; and			
10.2	Lifting appliance testing.			
11	**Corrosion and Preservation:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
11.1	Types of corrosion: 1. Dry; 2. Wet corrosion; 3. Single cell; 4. Selective phase; 5. Impingement attack; 6. Cavitation erosion; 7. Crevice corrosion; 8. Stress corrosion; and 9. Corrosion fatigue.			
11.2	Corrosion protection: 1. Improved design; 2. Drainage; 3. Accessibility; 4. Reduce protrusions; 5. Painting; 6. Galvanizing; 7. Metal spraying; 8. Non sacrificial coatings;			

Teaching Points		Instructional Strategies	Time (min)	References
	9. Plastics; and 10. QA.			
12	**Safety and Environmental Considerations**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
13	**Administration**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
13.1	Captain's Ship Book (CSB).			

5. Total EO Time: Total: 19 x 50 minute periods as follows:

   a. 18 x 50 min period – Classroom; and

   b. 1 x 50 min periods – EC (indicate total time or per student).

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 005 – MANAGE THE OPERATION OF HULL SYSTEMS

EO 005.02 DESCRIBE THE THEORY AND APPLICATION OF HEATING VENTILATION AND AIR CONDITIONING (HVAC) SYSTEMS

1. Performance: Describe the Theory and Application of Heating Ventilation and Air Conditioning (HVAC) Systems.

2. Conditions:

   a. Given:

      (1) Supervision; and

      (2) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) N/A.

3. Standard: The MS ENG shall demonstrate a thorough understanding of the theory and application of HVAC Systems.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Definition and Purpose**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
2	**History and Development**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
3	**Theoretical Overview and Terminology**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
3.1	Methods of Ventilation: 1. Natural; 2. Forced supply and natural exhaust;			

Teaching Points		Instructional Strategies	Time (min)	References
	3. Natural supply and Forced exhaust 4. Forced Supply and Forced exhaust 5. Auxiliary (portable)			
3.2	Ambient conditions 1. In-Theatre operational considerations			
3.3	Heat load requirements: 1. Machinery services; 2. Combat systems spaces; 3. Magazines; 4. Galley; and 5. Habitability spaces			
3.4	NBCD/Fire Fighting			
4	**Components**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
4.1	Chilled water system: 1. Ringmain; 2. Pumps; 3. Expansion tanks; 4. Heat exchangers; 5. Cooling medium; and 6. Control			
4.2	HVAC: 1. Ventilation fans/motors; 2. Intakes; 3. Filters (NBCD); 4. Dehumidification; 5. Heating coils; 6. Trunking; 7. Plenums; and 8. Control			
5	**Running Parameters and Protection**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		

Teaching Points		Instructional Strategies	Time (min)	References
6	**System Test and Trials**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
6.1	Citadel Test			
7	**Troubleshooting and Maintenance Considerations**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
7.1	Cooling medium			
7.2	Open - ship venting			
8	**Safety and Environmental Considerations**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
8.1	Bonding and Grounding			
8.2	High Voltages			
9	**In-Service Systems**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
9.1	HAL Class			
9.2	HDW Class			
9.3	PRO Class			
9.4	KIN Class			
9.5	VIC Class			

5. Total EO Time: Total: 5 x 50 minute periods as follows:

   a. 5 x 50 min period – Classroom;

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 005 – MANAGE THE OPERATION OF HULL SYSTEMS

EO 005.03 DESCRIBE THE THEORY AND APPLICATION OF POLUTION ABATEMENT SYSTEMS

1. Performance: Describe the Theory and Application of Pollution Abatement Systems.

2. Conditions:

   a. Given:

      (1) Supervision; and

      (2) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) N/A.

3. Standard: The MS ENG shall demonstrate a thorough understanding of the theory and application of Pollution Abatement Systems.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Definition and Purpose**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
2	**History and Development**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
3	**Theoretical Overview and Terminology:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
3.1	System requirements: 1. Collection;			

Teaching Points		Instructional Strategies	Time (min)	References
	2. Storage; and 3. Treatment.			
3.2	Collection Types: 1. Gravity; and 2. Vacuum.			
3.3	Treatment types: 1. Biological; 2. Electro-catalytic; 3. Chemical; and 4. Mechanical.			
4	**Components:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
4.1	Collection: 1. Toilets; 2. Drains; 3. Tanks; 4. Pumps; 5. Eductors; 6. Piping/vent; and 7. Overboard discharge.			
4.2	Treatment: 1. Pumps; 2. Macerators; 3. Book cell; 4. Settling tanks; 5. Piping/hoses; 6. Overboard discharge; and 7. Control.			
5	**Starting and stopping**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
6	**Running Parameters and Protection**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		

Teaching Points		Instructional Strategies	Time (min)	References
7	**Troubleshooting and Maintenance Considerations:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
7.1	foreign objects and debris; and			
7.2	system flush.			
8	**Corrosion and Preservation**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
9	**Safety and environmental considerations:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
9.1	Untreated discharge regulations;			
9.2	PPE;			
9.3	Hygiene; and			
9.4	Confined space procedures.			
10	**Administration:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
10.1	Class Certification of initial system.			
11	**In-Service systems:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
11.1	HAL Class;			
11.2	HDW Class;			
11.3	PRO Class;			
11.4	KIN Class; and			
11.5	VIC Class.			

5. Total EO Time: Total: 4 x 50 minute periods as follows:

   a. 3 x 50 min period – Classroom; and

b. 1 x 50 min period – Workshop demonstration;

6. Training and Learning Aids:

a. Static displays of Black and Grey water treatment equipment.

7. Remarks: Nil.

PO 005 – MANAGE THE OPERATION OF HULL SYSTEMS

EO 005.04 DESCRIBE THE THEORY AND APPLICATION OF SEA WATER SYSTEMS

1. Performance: Describe the Theory and Application of Sea Water Service Systems.

2. Conditions:

   a. Given:

      (1) Supervision; and

      (2) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) N/A.

3. Standard: The MS ENG shall demonstrate a thorough understanding of the theory and application of Sea Water Service Systems.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Definition and purposes**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
1.1	Fire Fighting			
1.2	Prewet			
1.3	Bilge pumping			
1.4	Ballast tanks			
1.5	Cooling			
1.6	Anchor wash			
2	**Theoretical Overview and Terminology**	**Method:** Lecture **Media:** PowerPoint		

Teaching Points		Instructional Strategies	Time (min)	References
		Environment: Classroom		
2.1	Number of outlets			
2.2	DC/NBCD/Fire Fighting			
2.3	Flexibility			
2.4	Galvanic corrosion			
3	**Components**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
3.1	Pumps: 1. Capacity			
3.2	Piping: 1. Risers; 2. Main; and 3. Jumpers			
3.3	Sea bays			
3.4	Eductors			
3.5	Valves: 1. Isolation; and 2. Regulating			
3.6	System redundancy			
4	**Materials**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
4.1	Copper/Nickel			
5	**Tests and Trials**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
5.1	Prewet			
5.2	Pump flow rates			
6	**Troubleshooting and Maintenance Considerations:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
6.1	Magazine sprays			

Teaching Points		Instructional Strategies	Time (min)	References
6.2	Cofferdams			
7	**Safety and Environmental Considerations**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
7.1	Ballast water management			
8	**In-Service systems**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
8.1	HAL Class			
8.2	HDW Class			
8.3	PRO Class			
8.4	KIN Class			
8.5	VIC Class			

5. Total EO Time: Total: 2 x 50 minute periods as follows:

   a. 2 x 50 min period – Classroom;

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 005 – MANAGE THE OPERATION OF HULL SYSTEMS

EO 005.05 DESCRIBE THE THEORY AND APPLICATION OF FRESH WATER DISTRIBUTION

1. Performance: Describe the Theory and Application of Fresh Water Distribution.

2. Conditions:

   a. Given:

      (1) Supervision; and

      (2) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) N/A.

3. Standard: The MS ENG shall demonstrate a thorough understanding of the theory and application of Fresh Water Distribution Systems.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Definition and Purpose**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
2	**Theoretical Overview and Terminology:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
2.1	Storage/location; and			
2.2	Treatment: 1. Chlorination; 2. Bromination; and 3. UV.			

Teaching Points		Instructional Strategies	Time (min)	References
2.3	Quality;			
3	**Components:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
3.1	Tanks;			
3.2	Pumps: 1. Capacity.			
3.3	Outlets;			
3.4	Treatment;			
3.5	Filling/transfer/supply mains;			
3.6	Ballast Water Tanks; and			
3.7	Calorifiers.			
4	**Troubleshooting and Maintenance Considerations:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
4.1	Water quality;			
4.2	System preservation;			
4.3	Super chlorination; and			
4.4	Tank inspections: 1. Tank hygiene.			
5	**Corrosion and Preservation:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
5.1	Non-toxic tank coatings.			
6	**Safety and Environmental Considerations**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
7	**In-Service systems**	**Method:** Lecture **Media:** PowerPoint		

Teaching Points		Instructional Strategies	Time (min)	References
		Environment: Classroom		
7.1	HAL Class;			
7.2	HDW Class;			
7.3	PRO Class;			
7.4	KIN Class; and			
7.5	VIC Class.			

5. Total EO Time: Total: 2 x 50 minute periods as follows:

   a. 2 x 50 min period – Classroom;

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 005 – MANAGE THE OPERATION OF HULL SYSTEMS

EO 005.06 DESCRIBE THE HULL SYSTEMS IN HMC SHIPS

1. Performance: Describe the Hull Systems in HMC Ships.

2. Conditions:

   a. Given:

      (1) Operational Ship;

      (2) Supervision; and

      (3) References;

   b. Denied:

      (4) Nil; and

   c. Environment:

      (5) All conditions, day and night.

3. Standard: The MS ENG shall demonstrate a thorough understanding of the operation of Hull Systems fitted on board.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Heating, Ventilation and Air Conditioning (HVAC):**	**Method:** OJT **Media:** N/A **Environment:** Ship		
1.1	NBC filtration units			
1.2	Air conditioning plenum			
1.3	Ship's citadels (narrative only)			
1.4	Smoke control (narrative only)			
1.5	KUs and FCUs (narrative only)			
2	**Pollution Abatement**	**Method:** OJT **Media:** N/A **Environment:** Ship		
2.1	Black water collection			
2.2	Grey water collection			

Teaching Points		Instructional Strategies	Time (min)	References
2.3	Sewage treatment			
3	**Sea Water Service system**	**Method:** OJT **Media:** N/A **Environment:** Ship		
3.1	Firemain: 1. Damage control; 2. Cooling services; and 3. Pumps			
3.2	Bilge systems			
4	**Fresh Water Storage and Distribution system**	**Method:** OJT **Media:** N/A **Environment:** Ship		

5. Total EO Time: Throughout the Afloat phase.

6. Training and Learning Aids: Nil.

7. Remarks: The MS ENG shall produce drawings that are:

   a. Neat/legible;

   b. Consistent with ship's fitted drawings; and

   c. Completer, to include as applicable:

      (1) Content and capacities;

      (2) Direction of flow;

      (3) Monitoring arrangements and parameters;

      (4) Control arrangements;

      (5) Valve types; and

      (6) A uniform colour-code.

PO 005 – MANAGE THE OPERATION OF HULL SYSTEMS

EO 005.07 DESCRIBE THE OPERATION OF HULL SYSTEMS IN HMC SHIPS

1. Performance: Describe the Operation of Hull Systems in HMC Ships.

2. Conditions:

   a. Given:

      (1) Operational Ship;

      (2) Supervision; and

      (3) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) All conditions, day and night.

3. Standard: The MS ENG shall demonstrate a thorough understanding of Hull Systems fitted on board.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**The MS ENG shall participate in all modes of operation (including flash up and shutdown) of hull systems and equipment fitted on board to include:**	**Method:** OJT **Media:** N/A **Environment:** Ship		
12	Heating, Ventilation and Air Conditioning (HVAC);			
1.2	Pollution Abatement;			
1.3	Seawater service system; and			
1.4	Fresh Water Storage and Distribution System.			

Teaching Points		Instructional Strategies	Time (min)	References
2	**The MS ENG shall participate in all modes of operation of the hull systems and equipment listed in para 10a to include:**	**Method:** OJT **Media:** N/A **Environment:** Ship		
2.1	Normal operation;			
2.2	Abnormal operations;			
2.3	Emergency operations;			
2.4	Operating parameters;			
2.5	Operational limitations;			
2.6	Monitoring methods;			
2.7	Flash ups;			
2.8	Shutdowns;			
2.9	Safety considerations;			
2.10	Troubleshooting; and			
2.11	Routine running maintenance.			

5. Total EO Time: Throughout the Afloat phase.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 006 – DIRECT EMERGENCY OPERATIONS

EO 006.01 DESCRIBE ENGINEERING EMERGENCY PROCEDURES

1. Performance: Describe Engineering Emergency Procedures.

2. Conditions:

   a. Given:

      (1) Supervision; and

      (2) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) N/A.

3. Standard: The MS ENG shall demonstrate a thorough understanding of the Engineering Emergency Procedures.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Engineering drills**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
1.1	Purposes			
1.2	Frequency			
1.3	Types			
1.4	Plan/scheduling			
1.5	Conduct 1. Reaction to an engineering emergency; and 2. Reaction to an engineering problem			
1.6	Communications			

Teaching Points		Instructional Strategies	Time (min)	References
1.7	Assessment			
2	**Command appreciation**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
3	**Operational limitations**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
4	**Critical Engineering Emergency drills**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
4.1	Main Lubricating Oil failure			
4.2	Steering			
4.3	Power Blackout at sea			

5. Total EO Time: Total: 3 x 50 minute periods as follows:

   a. 2 x 50 min period – Classroom; and

   b. 1 x 50 min periods – EC (indicate total time or per student).

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 006 – DIRECT EMERGENCY OPERATIONS

EO 006.02 DESCRIBE THE ENGINEERING EMERGENCY PROCEDURE IN HMC SHIPS

1. Performance: Describe Engineering Emergency Procedures in HMC Ships.

2. Conditions:

   a. Given:

      (1) Operational Ship;

      (2) Supervision; and

      (3) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) All conditions, day and night.

3. Standard: The MS ENG shall demonstrate a thorough understanding of the Engineering Emergency Procedures on board.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**The MS ENG shall participate in Engineering Drills:**	**Method:** OJT **Media:** N/A **Environment:** Ship		
1.1	As Engineering Roundsperson;			
1.2	As an Electrician of the Watch;			
1.3	As a Engineering Watchkeeper;			
1.4	As an Engineering Officer of the Watch; and			
1.5	Departmental drills planning: 1. Interdepartmental liaison; 2. Safety considerations;			

Teaching Points		Instructional Strategies	Time (min)	References
	3. Personnel requirements; and 4. Training requirements;			
2	**The MS ENG shall, to the satisfaction of the MSEO, produce a written report on the Engineering Drills to include:**	**Method:** OJT **Media:** N/A **Environment:** Ship		
2.1	Standard engineering watch response; and			
2.2	Operational limitations for all engineering emergencies and electrical power failure.			

5. Total EO Time: Throughout the Afloat phase.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 007 – MANAGE SHIP'S STABILITY

EO 007.01 PERFORM TRIGONOMETRIC FUNCTIONS

1. Performance: Perform Trigonometric Functions.

2. Conditions:

   a. Given:

      (1) Supervision; and

      (2) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) N/A.

3. Standard: The MS ENG shall demonstrate a thorough understanding of Trigonometric Functions.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Basic trigonometric functions**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
1.1	Right angle triangles			
1.2	Oblique angle triangles;			
1.3	Similar triangles; trigonometry			
1.4	Trigonometry functions to include: 1. Sine function; 2. Cosine function; and 3. Tangent function			
1.5	Radian measure			

5. Total EO Time: Total: 2 x 50 minute periods as follows:

   a. 2 x 50 min period – Classroom.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 007 – MANAGE SHIP'S STABILITY

EO 007.02 DEMONSTRATE AN UNDERSTANDING OF STABILITY TERMINOLOGY

1. Performance: Demonstrate an Understanding of Stability Terminology.

2. Conditions:

   a. Given:

      (1) Supervision; and

      (2) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) N/A.

3. Standard: The MS ENG shall demonstrate a thorough understanding of Stability Terminology.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Archimedes principle**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
2	**Density (fresh and salt water)**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
3	**Specific volume**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		

Teaching Points		Instructional Strategies	Time (min)	References
4	**Specific gravity**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
5	**Displacement**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
6	**Force**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
7	**Couple**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
8	**Moments**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
9	**Moments of inertia**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
10	**Floatation**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
11	**Buoyancy**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		

5. Total EO Time: Total: 3 x 50 minute periods as follows:

   a. 3 x 50 min period – Classroom.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 007 – MANAGE SHIP'S STABILITY

EO 007.03 EXPLAIN THE PRINCIPLES OF STABILITY

1. Performance: Explain the Principles of Stability.

2. Conditions:

   a. Given:

      (1) Supervision; and

      (2) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) N/A.

3. Standard: The MS ENG shall demonstrate a thorough understanding of Stability.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Terminology**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
1.1	Stability			
1.2	Force of gravity			
1.3	Force of buoyancy			
1.4	Centre of gravity			
1.5	Centre of buoyancy			
1.6	Metacentre			
1.7	Metacentric height			
1.8	Height of metacentre			
1.9	Righting moments			
1.10	Righting arm			
1.11	Upsetting moments			
1.12	Upsetting arms			

Teaching Points		Instructional Strategies	Time (min)	References
1.13	Trim			
1.14	List			
1.15	Loll			
1.16	Heel			
2	**States of stability**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
2.1	States of equilibrium: 1. Stable; 2. Neutral; and 3. Unstable			
2.2	Period of roll			
2.3	Transverse stability			
2.4	Longitudinal stability			
2.5	Effects of GMs			
2.6	Vertical weight shifts			
2.7	Transverse weight shifts			
2.8	Free surface and pocketing			
2.9	Free communication			
2.10	Countermeasures to improve stability			
2.11	Countermeasures to improve stability			
3	**Inclining experiment**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
3.1	Purpose			
3.2	Purpose			
4	**Conditions**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
4.1	Dived displacement			
4.2	Submerged standard condition			
4.3	Normal surfaced condition			
4.4	Full buoyancy condition			

Teaching Points		Instructional Strategies	Time (min)	References
4.5	Manoeuvring limitation diagram			
4.6	Trim polygon			

5. Total EO Time: Total: 29 x 50 minute periods as follows:

   a. 29 x 50 min period – Classroom.

6. Training and Learning Aids:

   a. Films and ship model.

7. Remarks: Nil.

PO 007 – MANAGE SHIP'S STABILITY

EO 007.04 EXPLAIN STABILITY CURVES

1. Performance: Explain Stability Curves.

2. Conditions:

   a. Given:

      (1) Supervision; and

      (2) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) N/A.

3. Standard: The MS ENG shall demonstrate a thorough understanding of Stability Curves.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Terminology:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
1.1	Perpendiculars forward and aft;			
1.2	Length between perpendiculars;			
1.3	Length between draft;			
1.4	Longitudinal centre of flotation; and			
1.5	Mean draft.			
2	**Static Stability Curve:**	**Method:** Lecture **Media:** PowerPoint		

Teaching Points		Instructional Strategies	Time (min)	References
		Environment: Classroom		
2.1	Tender;			
2.2	Stiff;			
2.3	Range of stability;			
2.4	Point of vanishing stability;			
2.5	Maximum righting arms;			
2.6	Dynamic stability;			
2.7	Positive and negative GM;			
2.8	Angle of loll;			
2.9	Righting arms; and			
2.10	Angle of heel.			
3	**Hydrostatic Curves:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
3.1	Displacement;			
3.2	Vertical centre of buoyancy;			
3.3	Transverse metacentre;			
3.4	Tons per inch/centimetre immersion;			
3.5	Moment to change trim;			
3.6	Longitudinal centre of buoyancy; and			
3.7	Longitudinal centre of flotation.			
4	**Cross Curves of Stability**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		

5. Total EO Time: Total: 5 x 50 minute periods as follows:

   a. 5 x 50 min period – Classroom.

6. Training and Learning Aids:

a. Films.

7. Remarks: Nil.

PO 007 – MANAGE SHIP'S STABILITY

EO 007.05 USE THE MANUAL OF TRIM AND STABILITY

1. Performance: Use the Manual of Trim and Stability.

2. Conditions:

   a. Given:

      (1) Supervision; and

      (2) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) N/A.

3. Standard: The MS ENG shall demonstrate a thorough understanding of the Manual of Trim and Stability.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Vessel particulars**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
2	**Instruction for the safe working of the ship**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
3	**Stability requirement limits**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
4	**Intact stability conditions**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		

Teaching Points		Instructional Strategies	Time (min)	References
5	**Damage stability conditions**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
6	**Stability calculation procedures**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
6.1	Intact			
6.2	Damage			
7	**Tanks particulars**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
8	**Icing particulars**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
9	**Hydrostatic data**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		

5. Total EO Time: Total: 10 x 50 minute periods as follows:

   a. 8 x 50 min period – Classroom; and

   b. 2 x 50 min periods – EC (indicate total time or per student).

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 007 – MANAGE SHIP'S STABILITY

EO 007.06 SOLVE PROBLEMS RELATED TO SHIP'S STABILITY

1. Performance: Solve Problems Related to Ship's Stability.

2. Conditions:

   a. Given:

      (1) Supervision; and

      (2) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) N/A.

3. Standard: The MS ENG shall demonstrate a thorough understanding of Ship's Stability problem solving.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Stability mathematical calculations:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
1.1	Metacentric height;			
1.2	Height of metacentre;			
1.3	Displacement;			
1.4	Period of roll;			
1.5	Vertical weight shift;			
1.6	Transverse weight shift;			
1.7	Diagonal weight shift;			
1.8	Addition/removal of weight;			
1.9	Free surface and pocketing;			
1.10	Free communication;			

Teaching Points		Instructional Strategies	Time (min)	References
1.11	Mean draft;			
1.12	Trim change;			
1.13	Parallel sinkage/riseage;			
1.14	Flooding rate through a hole;			
1.15	Docking/grounding;			
1.16	Docking grounding force; and			
1.17	Line pull;			
2	**Icing:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
2.1	Type of ice;			
2.2	Factors affecting ice formation on ships;			
2.3	Effect on ship and on stability; and			
2.4	Action to reduce ice formation;			
3	**Grounding:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
3.1	Type of bottom;			
3.2	Effect on ship's stability; and			
3.3	Action to be taken before/after grounding; and			
4	**Effects of docking on ship's stability**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		

5. Total EO Time: Total: 23 x 50 minute periods as follows:

   a. 21 x 50 min period – Classroom; and

   b. 2 x 50 min periods – EC (indicate total time or per student).

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 008 – DIRECT DAMAGE CONTROL OPERATIONS

EO 008.01 DESCRIBE THE DUTIES OF SHIP'S DAMAGE CONTROL OFFICER (DCO)

1. Performance: Describe the Duties of the Ship's Damage Control Officer (DCO).

2. Conditions:

   a. Given:

      (1) Supervision; and

      (2) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) N/A.

3. Standard: The MS ENG shall demonstrate a thorough understanding of the duties of the Ship's Damage Control Officer (DCO) during Damage Control activities.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Damage Control Organization**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
1.1	Positions			
1.2	Duties and responsibilities of each position			
2	**Assess damaged state**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
3	**Assess effectiveness of initial action**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		

Teaching Points		Instructional Strategies	Time (min)	References
4	**Determine the priorities for damage control**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
5	**Direct remedial action**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
6	**Provide advice to Command to include**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
6.1	Impact of damage on trim and stability			
6.2	Impact of damage outside the main machinery spaces on ship's propulsion plant operation			
6.3	Requirement to flood or spray magazines			
6.4	Impact of damage on electrical PG&D systems. 1. Requirement to activate fire suppression system; and 2. Consideration for depth requirements			

5. Total EO Time: Total: 47 x 50 minute periods as follows:

   a. 36 x 50 min period – Classroom; and

   b. 11 x 50 min periods – EC (indicate total time or per student).

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 008 – DIRECT DAMAGE CONTROL OPERATIONS

EO 008.02 DESCRIBE DAMAGE CONTROL OPERATIONS IN HMC SHIPS

1. Performance: Describe Damage Control Operations in HMC Ships.

2. Conditions:

   a. Given:

      (1) Operational Ship;

      (2) Supervision; and

      (3) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) All conditions, day and night.

3. Standard: The MS ENG shall demonstrate a thorough understanding of Damage Control Operations on board.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**The MS ENG shall participate in Damage Control exercises as:**	**Method:** OJT **Media:** N/A **Environment:** Ship		
1.1	Section base communicator/plotter (70% of DC exercises);			
1.2	Observe the Command Team (including Rapid Response) on the Bridge and Operations Room (10% of DC exercises);			
1.3	Observe Section Base 3 during Emergency Flying Operations (5% of DC exercises); and			

Teaching Points		Instructional Strategies	Time (min)	References
1.4	Damage Control Officer (DCO) in HQ1 (15% of DC exercises).			
2	**The MS ENG shall produce a written report on Damage Control Organization to include:**	**Method:** OJT **Media:** N/A **Environment:** Ship		
2.1	Roles and responsibilities of Damage Control Organization: 1. Command; 2. HQ1/HQ2; 3. Section Base Teams (SBTs); 4. Manning pool; 5. Emergency Response Team (ERT); and 6. Casualty Clearing Team (CCT).			
2.2	Damage Control Exercises: 1. Purpose; 2. Frequency; 3. Types; 4. Planning and Scheduling; 5. Monitoring and assessing; 6. Command appreciation; and 7. Safety considerations.			

5. Total EO Time: Throughout the Afloat phase.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 009 – MANAGE SHIPBOARD LIQUIDS

EO 009.01 DESCRIBE SHIPBOARD LIQUIDS

1. Performance: Describe Shipboard Liquid Handling.

2. Conditions:

   a. Given:

      (1) Supervision; and

      (2) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) N/A.

3. Standard: The MS ENG shall demonstrate a thorough understanding of Shipboard Liquid Handling.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Definition and Purpose**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
2	**Classification systems**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
3	**POL characteristics and terminology**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
3.1	American Petroleum Institute (API)			
3.2	Specific gravity			
3.3	Flash point			

Teaching Points		Instructional Strategies	Time (min)	References
3.4	Physical properties of lubricating oils			
3.5	Characteristics/identification of fuels, lubricants and their substitutes (alternate and emergency); and			
3.6	Effect of temperature on volume of liquids			
4	**POL types and their storage**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
4.1	Fuels: 1. NATO F-76 (DFO); 2. NATO F-44 (JP4/5); and 3. Gasoline			
4.2	Lubricants: 1. NATO O-240 Naval Steam Turbine and Main Gearing Lubricating Oil (CGSB 3GP-357); 2. NATO O-156 (MIL-PRF-2399 Synthetic Lubricating Oil); 3. NATO H-576 (CGSB 3GP-36M); and 4. NATO O-1236 (SAE 15W-40);			
4.3	Greases			
5	**Thermal Fluid**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
6	**Testing**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
6.1	Types of tests: 1. Visual; 2. Colour;			

Teaching Points		Instructional Strategies	Time (min)	References
	3. Guerin test kit; 4. Water test; 5. Bottom water and sediment test; 6. Flash point test; 7. Exxon hydro; 8. Clear and bright; 9. DAND samples; 10. OCCAP; and 11. Kittiwake			
7	**Contamination**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
7.1	Water			
7.2	Microbiological Growth (MBG)			
7.3	Surfactants			
7.4	Solids			
8	**Liquid handling policies and procedures**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
8.1	Cleaning and gas-freeing of on board liquid storage tanks			
8.2	Policies, procedures, and practices for the safe handling of fluids			
8.3	Fuel management and accounting: 1. Daily liquid reports; 2. Fuel tank rotation; and 3. Polishing			
9	**Liquid handling operations**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
9.1	Stability considerations			
9.2	Safety precautions during fuelling			

Teaching Points		Instructional Strategies	Time (min)	References
9.3	Replenishment underway and alongside: 1. Pre-fuelling requirements; 2. RAS methods			
9.4	Hazards of shipboard fuel handling			
10	**Environmental considerations**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
10.1	NAVORDS			
10.2	EMS			
11	**Administration**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
11.1	Procurement policies			

5. Total EO Time: Total: 8 x 50 minute periods as follows:

   a. 7 x 50 min period – Classroom; and

   b. 1 x 50 min periods – EC (indicate total time or per student).

6. Training and Learning Aids:

   a. Test kits.

7. Remarks: Nil.

PO 009 – MANAGE SHIPBOARD LIQUIDS

EO 009.02 MANAGE SHIPBOARD LIQUIDS

1. Performance: Describe Shipboard Liquid Handling in HMC Ships.

2. Conditions:

   a. Given:

      (1) Operational Ship;

      (2) Supervision; and

      (3) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) All conditions, day and night.

3. Standard: The MS ENG shall demonstrate a thorough understanding of Shipboard Liquid Handling Operations on board.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**The MS ENG shall participate in liquid handling operations and advise command to include:**	**Method:** OJT **Media:** N/A **Environment:** Ship		
1.1	Replenishment underway and alongside: 1. Understudy the Upper Deck Engineer; 2. Understudy the Flight Deck Engineer; 3. Prepare fuelling orders; 4. Conduct fuelling brief; 5. Observe upper deck operations; 6. Observe Command on the bridge;			

Teaching Points		Instructional Strategies	Time (min)	References
	7. Understudy the Fuelling Communicators; 8. Understudy the Fuelling IC; and 9. Understudy the MSEO (Fuelling Officer).			
1.2	Liquid husbandry underway and alongside: 1. Fuel Transfer; 2. FW/SW/Ballasting Operations; 3. Embarkation and storage of POLs; 4. Understudy the Upper Deck Engineer; and 5. Understudy the Aircraft Support Technician			
2	**The MS ENG shall produce a written report on Shipboard Liquid Handling Operations to include:**	**Method:** OJT **Media:** N/A **Environment:** Ship		
2.1	Types and quantity of liquids held on board;			
2.2	Stability and operational considerations;			
2.3	Safe handling, stowage and transfer;			
2.4	Environmental considerations;			
2.5	Duties and responsibilities of key personnel during replenishment at sea and alongside;			
2.6	Liquid quality assessment;			
2.7	Response to a POL spill;			
2.8	Replenishment at sea equipment including PRO Class RAS capability; and			
2.9	Purpose of fuel records and reporting.			

5. Total EO Time: Throughout the Afloat training phase.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 010 – MANAGE DOCKING OPERATIONS

EO 010.01 DESCRIBE DOCKING OPERATIONS

1. Performance: Describe Docking Operations.

2. Conditions:

   a. Given:

      (1) Supervision; and

      (2) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) N/A.

3. Standard: The MS ENG shall demonstrate a thorough understanding of Docking Operations.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Docking Policies and Principles**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
1.1	Purposes: Cleaning and painting the ship's bottom; Examination and repair of underwater fittings; and Measurement of stern tube, rudder and A-bracket bearing clearances;			
1.2	Terminology			
1.3	Frequency: Periodic; Unscheduled; and			

Teaching Points		Instructional Strategies	Time (min)	References
	Emergency			
1.4	Types: Types of docks: Floating; Graving; Synchro-lift; and Marine railway; Manned/unmanned dockings: Care and Custody of Vessel Certificate (unmanned only)			
1.5	Organizations involved and their responsibilities: 1. NDQAR; 2. Formation headquarters; 3. Docking Officer; 4. FMF; 5. Dockmaster; and 6. Ship			
1.6	Facilities: 1. Home port: 1. Government docks; and 2. Private industry docks. 2. Foreign			
1.7	Equipment considerations: 1. Firemain: 1. Transfer; 2. Supply pressure; and 3. Equipment cooling; 2. Power; and 3. Other typical alongside connections 4. Controlled dock access			
2	**Docking Procedures**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
2.1	Stability requirements			
2.2	Safety:			

Teaching Points		Instructional Strategies	Time (min)	References
	1. Personnel; 2. Equipment; and 3. Maintenance of exit routes			
2.3	Documentation 1. Docking memoranda; 2. Weight log; and 3. Report of docking			
2.4	Docking plan: 1. Ship's docking marks drawings			
2.5	Docking preparations: 1. De-ammunitioning; 2. De-fuelling; 3. De-storing; 4. Heel and trim; 5. Docking marks; 6. Locked rudders; 7. Underwater hull equipment (Sonar, etc.); 8. Docking calculations; 9. Dock inspections; and 10. Reports from other departments			
2.6	Interdepartmental considerations: 1. MSEO's responsibilities; 2. CSEO's responsibilities; 3. SYO's responsibilities; and 4. DeckO's responsibilities			

5. Total EO Time: Total: 8 x 50 minute periods as follows:

   a. 4 x 50 min period – Classroom;

   b. 3 x 50 min period – Dockyard tour;

   c. 1 x 50 min periods – EC (indicate total time or per student).

6. Training and Learning Aids: Nil.

7. Remarks:

a. The term docking in this EO refers to both docking and undocking operations.

b. If possible, this EO should be instructed by the FMF Naval Architecture Officer, Docking Officer, or another similarly-qualified individual.

PO 011 – ADMINISTER RISK MANAGEMENT PROGRAMS

EO 011.01 DESCRIBE SHIPBOARD SAFETY AND ENVIRONMENTAL PROCEDURES

1. Performance: Describe Shipboard Safety and Environmental Procedures.

2. Conditions:

   a. Given:

      (1) Supervision; and

      (2) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) N/A.

3. Standard: The MS ENG shall demonstrate a thorough understanding of Shipboard Safety and Environmental Procedures.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**The purpose, policy, procedure and MSEO managerial responsibilities for the following on the board safety activities:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
1.1	Gas freeing and certification;			
1.2	Burning and welding;			
1.3	Confined space entry;			
1.4	Diving Operations;			
1.5	Man aloft: 1. RADHAZ (non-ionizing); and 2. Ionizing radiation;			
1.6	Ammunitioning.			

Teaching Points		Instructional Strategies	Time (min)	References
2	**The purpose, policy, procedure and MSEO managerial responsibilities for the following environmental activities:**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom		
2.1	POL spill response;			
2.2	Halocarbon / Ozone Depleting Substance (ODS) release;			
2.3	Liquid Effluent management;			
2.4	Waste management; and			
2.5	Hazardous materials.			

5. Total EO Time: Total: 4 x 50 minute periods as follows:

    a. 3 x 50 min period – Classroom; and

    b. 1 x 50 min periods – EC (indicate total time or per student).

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

PO 011 – ADMINISTER RISK MANAGEMENT PROGRAMS

EO 011.02 DESCRIBE RISK MANAGEMENT PROGRAMS IN HMC SHIPS

1. Performance: Describe Risk Management Programs in HMC Ships.

2. Conditions:

   a. Given:

      (1) Operational Ship;

      (2) Supervision; and

      (3) References;

   b. Denied:

      (1) Nil; and

   c. Environment:

      (1) All conditions, day and night.

3. Standard: The MS ENG shall demonstrate a thorough understanding of Risk Management Programs on board.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**The MS ENG shall participate in the environmental protection program to include:**	**Method:** OJT **Media:** N/A **Environment:** Ship		
1.1	Ship's environmental audit;			
1.2	Review of Halocarbon documentation;			
1.3	Halocarbon leak detection;			
1.4	Halocarbon release message;			
1.5	Handling, storage and disposal of hazardous material; and			

Teaching Points		Instructional Strategies	Time (min)	References
1.6	Ship's environmental committee meeting.			
2	**The MS ENG shall participate in the general safety program to include:**	**Method:** OJT **Media:** N/A **Environment:** Ship		
2.1	Ship's general safety audit or SPDET; and			
2.2	Ship's general safety committee meeting.			
3	**The MS ENG shall produce a written report on the conduct of the safety procedures to include:**	**Method:** OJT **Media:** N/A **Environment:** Ship		
3.1	Gas freeing and certification;			
3.2	Burning and welding;			
3.3	Confined space entry;			
3.4	Diving operations; and			
3.5	Man aloft: 1. RADHAZ (non-ionizing); and 2. Ionizing radiation.			
3.6	Ammunitioning.			

5. Total EO Time: Throughout the Afloat phase.

6. Training and Learning Aids: Nil.

7. Remarks: Nil.

QUALIFICATION STANDARD ANNEXES

ANNEX A – MAIN REFERENCES

1. This annex lists all references used in this QS and TP.

   a. The previous version of the QSP did not have this annex filled out. This will be created at a later date.

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A		**Official CAF Publications and Directives that relate to this Qualification**
A01		
A02		
A03		
A04		
B		**(Non-CAF) Government of Canada Publications & Foreign Military (Non-DND) Publications that relate to this Qualification**
B01		
B02		
B03		
C		**Civilian Publications and Manuals that relate to this Qualification**
C01		
C02		
D		**Other Publications and Manuals that relate to this Qualification**
D01		
D02		

NOTE: If additional references are identified during the TPWB this annex can be updated secretarially through NPTG HQ.

ANNEX B – TASK LIST

1. The task list includes all of the tasks associated with this qualification, and the outcome of the Train / No-train decisions. If a task is designated as requiring training, the PO where it is covered is indicated. If a task is designated as not requiring training, the justification for that designation is provided.

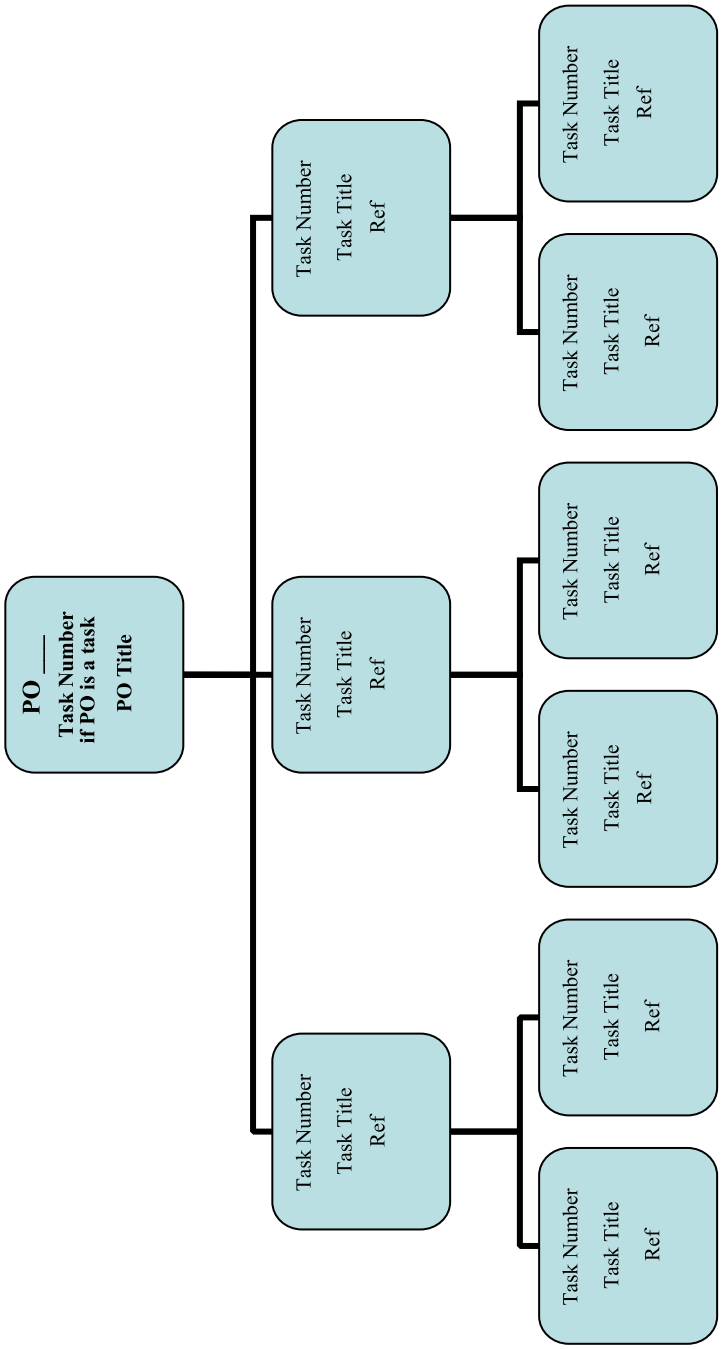
2. The following table defines some of the criteria used to support a no-train designation in the task list. Other reasons may be used if required.

   a. The previous version of the QSP did not have this annex filled out. This will be created at a later date.

CRITERIA	DEFINITION
A	**Previously qualified.** Tasks and knowledge already possessed by the student by CAF and/or RCN courses, training and/or experience.
B	**Will receive on future courses.** Tasks and knowledge will be covered as part of other CAF and/or RCN courses and/or training.
C	**Difficulty.** Task is not difficult to learn; formal training is not required.
D	**Importance.** Task is not important; it has no significant impact on mission success.
E	**Frequency.** Task is rarely performed, no formal training is required.
F	**Lack of equipment.** Currently there is no equipment in the CAF inventory to train this task.

Task #	Task Statement	Train / No Train	PO #	Substantiation/Comments

ANNEX C – QS SCALAR



ANNEX D – RECORD OF DECISIONS QSWB

1. This record summarizes the conduct of the QSWB including major activities, important discussions and contentious issues, as well as all decisions and recommendations made by the Board. A description of the reasoning which supports a decision or recommendation is provided when required.

Title	**Marine System Engineering Basic Officer Qualification QSPWB**			
Location of Board	**CFNES Halifax**			
Dates	**Part 1: 25 Feb to 22 Mar 02** **Part 2: 27 Jan to 21 Feb 03.**			
Board Composition				
	Surname	Initials	Rank	Unit
Chairperson (Part 1)	McDonald	D	LCdr	CFNESH
Chairperson (Part 2)	Bonnah	M	LCdr	CFMWC OTEG
DMTE Rep	Swann	D	Lt(N)	DMTE 2-4-2
TDO	Layden	F	Lt(N)	CFNESH QAO
MARL SME (Part 1)	Gilson	G	Lt(N)	CFNESH
MARL SME (Part 2)	Dunkley	E	Lt(N)	CFNESH
MARP SME (Part 1)	McIssac	J	LCdr	CFFSE
MARP SME (Part 2)	Gauthier	R	LCdr	CANFLTPACH Q
Secretary (Part 1)	Pare	S	Lt(N)	CFNESH
Secretary (Part 2)	Lawton	J	Lt(N)	CFNESH

Boards' Objective:

To write the MS ENG QSP. To achieve this objective, the MS ENG Job Based Occupation Specification (JBOS) was used as the primary document. Specifically, the QSPWBs' work was to:

a. Review and propose amendments to the JBOS;
b. Conduct the Task Analysis (Train/No Train/Delete decisions);
c. Create the Task Scalar;
d. Develop the Performance Objectives list;
e. Develop the Lesson Sepcifications list;
f. Identify the training aids requirements;
g. Develop the reference list;
h. Determine the Assessment criteria (Enabling Checks and Performance Checks) from
which QSP Chapter 3 could be written;
i. Estimate the course timings from which a pilot course serial duration could be calculated;
j. Complete the QSP Chapter 1 Course Management Details; and
k. Complete the QSP Annexes.

Task Serial	Task Element	Substantiation
AT0005	Prepare maritime technical policy documents (eg. MARCORDs, FTBs)	Low frequency task, easily learned on the job
AT0010	Consult with foreign governments on the sale/purchase of naval systems	Low percent performing (<30%)
AT0015	Consult with foreign governments on the design of naval systems	Low percent performing (<30%)
AT0020	Exchange information/technology with national and international agencies	Low percent performing (<30%)
AT0025	Arrange for the provision of naval resources to other government departments	Low percent performing (<30%)
AT0035	Prepare inputs to defence capital equipment procurement plans	Low frequency task, easily learned on the job
AT0050	Prepare energy conservation proposals	Low frequency task, easily learned on the job
AT0055	Approve amendments to technical manuals, publications, orders and regulations	Not required at this level
Etc…		

RECORD OF AMENDMENT & AMPLIFICATION

1. The record of amendment and amplification will be filled in whenever a change to the QS is initiated. This section is used to provide amplification details and provide a history of the changes made. In the requirement for the amendment section, the member inputting the changes will transpose the information from the submitted QS Change Request (QSCR), thus ensuring the QSP has the most relevant information.

Version:	**2.0**
Date: 1 July 2023	**Who Initiated Change: HMCS Venture**
Requirement for the Amendment: New format of RCN QSP	
Description of Changes: The QSP was rewritten to the new format of the RCN QSP. There was a removal of POs 12 to 20, as they were moved to Naval Engineering Common in 2012, with the old version of the QSP this rewrite was based off of was dated 2003. There were no newer QSPs found.	
Change Tracking number:	
Trg Mgr: Lt(N) Newcombe	

TRAINING PLAN ANNEXES

ANNEX E – RESOURCES, EQUIPMENT, SUPPLIES, AND ANCILLARIES

1. The following resources, equipment, supplies, and other elements are required in support of training. Resources are described in terms of formed organizations and are to arrive equipped with all personnel, vehicles, weapons, communications, and ancillary equipment required for conduct and integral support of operations described in the POs.

a. This information was not found in the previous version of the QSP. In the rewrite this was not filled out however this will be required to be filled out in the future.

2. Facilities/Trainers

SER	DESCRIPTION		QTY	REMARKS
a.				
b.				
c.				
NOTES:				

3. Ammunition

SER	LOG GUIDE	DESCRIPTION		QTY	REMARKS
a.					
b.					
c.					
NOTES:					

4. Weapons

SER	NSN	DESCRIPTION		QTY	REMARKS
a.					
b.					
c.					
NOTES:					

5. Specialized/Technical Equipment

SER	NSN	DESCRIPTION		QTY	REMARKS
a.					
b.					
c.					
NOTES:					

6. Transport/Vehicles

SER		NSN	DESCRIPTION		QTY	REMARKS
a.						
b.						
c.						
	NOTES:					

7. Communication Equipment/Frequencies

SER	NSN	DESCRIPTION		QTY	REMARKS
a.					
b.					
c.					
NOTES:					

8. Training Aids

SER	DESCRIPTION		QTY	REMARKS
a.				
b.				
c.				
NOTES:				

9. POL Products

SER	NSN	DESCRIPTION		QTY	REMARKS
a.					
b.					
c.					

NOTES:		

10. General Purpose Stores

SER	DESCRIPTION		QTY	REMARKS
a.				
b.				
c.				

NOTES:		

11. Stationery

SER	DESCRIPTION		QTY	REMARKS
a.				
b.				
c.				

NOTES:		

12. Personal Equipment

All students will have in their possession the personal equipment detailed in the course joining instructions. All personal equipment to be assigned to the students are accounted for in the resource tables above.

ANNEX F – COSTING INFORMATION

This information was not found in the previous version of the QSP. In the rewrite this was not filled out however this will be required to be filled out in the future.

Estimates only, based on CF Cost Factors Manual

cost-factors-manual (mil.ca)

****5 year total based on one course per year INCLUDING Initial Set-Up. Costs may be adjusted accordingly.**

Course: EXAMPLE		Serial:				
TD	No. Of Students / instructors /	Cost per each	# Days (if applicable)	Initial TE Set-up	Each successive course	5 Year total
Subtotal						
Material	Quantity	Cost per each	# Days (enter 1 if not applicable)			
PP&S (Estimate)						
Classroom (square meters)					…	
Computer requirements						
IT Equip (Projector, Smart board etc.)						
DWAN connection (Set up)						
Link Toolkit (on DVD)						
Instructional Materials (Printed / Handouts etc.)						
Subtotal						
PYs (only if course includes incremental on SWE, or course to be delivered off site	Quantity	Cost per day* (SWE if necessary)	# Days (if applicable)			
	Total Estimated Costs					

ANNEX G – SUGGESTED SEQUENCING AND TIMINGS

1. This annex is to be used to properly schedule all required classrooms, trainers, ships, personnel, etc. for the duration of the course. Admin and PT periods are accounted for IAW NPTORD 4500-22 Annex B (QSP Instructions).

2. The tables are completed in the preferred sequence for lessons and resources. Special instructions to schedulers shall be included as "Notes" following the applicable PO or Module.

3. The table below is available to use as a template for calculations.

4. The below table is filled out with the information filled out in the previous QSP. A more comprehensive review of timings will be conducted at a later date.

Allocation of ATT	**Minutes**
Actual Training Time	
ATT Sub-total	
Allocation of BAT, TAT, TTT, PT (Course > 31 days)	
Base Administrative Time (BAT)	
Training Administrative Time (TAT)	
Training Travel Time (TTT)	
Physical Fitness (PT)	
Sub-total	
Course Duration Calculation (ATT+BAT+TTT+TAT+PT) **Total mins/400 = # days** (rounded up)	
Total Training Days	**110**

OPTIMUM TRAINING SCHEDULE

Week 1 **Time**	**Day 1**	**Day 2**	**Day 3**	**Day4**	**Day 5**
Period 1 **0800-0850**	**EO/EC /PC/Admin** Title(s) / Activity				
Period 2 **0900-0950**	**EO/EC /PC/Admin** Title(s) / Activity				
Period 3 **1000-1050**	**EO/EC /PC/Admin** Title(s) / Activity				
Period 4 **1100-1150**	**EO/EC /PC/Admin** Title(s) / Activity				
Period 5 **1300-1350**	**EO/EC /PC/Admin** Title(s) / Activity				
Period 6 **1400-1450**	**EO/EC /PC/Admin** Title(s) / Activity				
Period 7 **1500-1550**	**EO/EC /PC/Admin** Title(s) / Activity				
Period 8 **1600-1650**	**EO/EC /PC/Admin** Title(s) / Activity				

ANNEX H – ASSESSMENT PLAN

This annex was not found in the previous QSP. It is left here to be filled out at a later date.

EC / PC	Scope	Purpose	How	When	Time	Resources	Limitations	Pass/Fail Criteria	Impact
Entire QSP	PC 001 MS BOQ Qualification Board	To evaluate the candidate's technical ability to perform as a Naval Combat Systems Engineer.	Qualification Board (oral board)	On completion of OJPR and NEC OJPR	Up to 300 min, plus up to 60 min prep time	Boardroom. Board Chair (Cdr. NAV ENG Cdr) Examining Officer (LCdr MS ENG) Board Member (Lt(N) MS ENG)	No textbooks. No notes. No help. May be stopped if candidate does not achieve at least 60% on board questions 1 and 2.	Pass mark: The candidate shall achieve a minimum overall average mark of 60%, including a minimum of 60% for each of questions 1 and 2 (critical questions).	Trainees will require a re-board. TRB upon failure of reboard.

ANNEX I – ASSESSMENT CHECKLISTS

The MS BOQ board is a technical board. Knowledge of key systems is expected from the candidate. The purpose of this qualification board is not simply to assess technical competency, but the ability to assess, analyze and convey complex technical information verbally, with comprehension beyond memorization. The candidate is expected to memorize key technical parameters and operating characteristics. The candidate shall demonstrate the ability to stand and deliver, conveying engineering fundamentals, technical and military knowledge in a clear, concise, and professional manner.

Technical Efficacy, Demonstrating Technical Knowledge, Technical Currency, Policy / Program Application and being a Technical Advisor are all facets of PaCE for all occupations and shall be assessed by the board.

The board consists of 5 questions total, with the following recommended schedule and timings,

Q	Portion	Time (mins)	Morning Board 0800 Start	Afternoon Board 1300 Start
	Candidate Major System Drawing	60	0700-0800	1200-1300
	Candidate and Board Member Introduction	10	0800-0810	1300-1310
1	Major System	60	0810-0910	1310-1410
2	Engineering Drills	15	0910-0925	1410-1425
	Break	20	0925-0945	1425-1445
3	Minor System or Damage Control System	25	0945-1010	1445-1510
4	Stability	30	1010-1040	1510-1540
5	Divisional Administration	20	1040-1100	1540-1600
	Board Deliberation / Marking	15	1100-1115	1600-1615
	Initial Feedback and Completion	5	1115-1120	1615-1620

Question 1 | Major System

60 minutes preparation; 45 minutes delivery, 15 minutes follow-on questions

The candidate shall create, from memory, a detailed and well laid out drawing of the system as fitted on their ship. The drawing shall be neat, clear and of sufficient detail to explain the system's operation. The candidate should use supplemental drawings for major components, and use an appropriate colour, and scale. The candidate will be marked on the following topics.

- **Purpose.** What is the Design Intent of the system and how does the system contribute to float, move, and fight.

- **Theory**. The candidate shall provide some background on the physics and engineering fundamentals of the system. Include supplemental sketches of the principles of operation, engineering application, building on their post secondary education, NEI, SEMC and MSE Applications training.

- **Overview.** Provide a general overview of the system and orient the drawing. What spaces is the system in? Is there a port and starboard, primary, and secondary system? Was this part of an upgrade, refit, or as fitted when commissioned? Is this system the same for all ships within class? What colors are used in the drawing and why, and what legend and symbology is used? Standard symbols for pumps, valves, orifice plates, heat exchangers, reservoirs, sumps, etc. shall be used.

- **Major Components.** List and discuss at a minimum, the operation, location, function, and characteristics of all components, building on the theory. Discuss each component's capabilities, limitations, parameters, operating ranges, alarms, and trips.

- **Functional Description.** Describe the normal operation of the complete system. A suggestion is to trace out the flow path for fluid and electrical current. The functional description should start with an operator at a console; entering a value or change in state and end with actuation, feedback, and monitoring of the system.

- **Modes of Operation**. Is it always in operation, only at sea, or only in specific drive modes or missions? Primary and alternate operation modes shall be discussed for systems with local, remote, or primary, backup and emergency operation modes.

- **Preventative Maintenance and Corrective Maintenance routines.** All normal routine preventative and common corrective maintenance shall be discussed. The candidate is not expected to know specific routine frequencies or a complete list of maintenance and repair procedures but should know what is involved in common maintenance operations.

- **Trials, specific operating considerations.** The candidate should be aware of the relevant trials and trial programs for major systems. They are not expected to know the detailed trial agendas or frequency, however a general awareness of who is involved, the duration of the trial, as well as operational and staffing considerations are expected. On questioning, the candidate should be able to briefly discuss operational limitations, technical bulletins and fleet advisories and general directives on best practices, if relevant.

- **Control & Monitoring.** Can this system be controlled and/or monitored remotely, and if so, what aspects can/cannot?

- **Interdependencies.** All systems rely on other systems for input, output and interoperability. The relationship, reliance, and resiliency of the topic system to other systems shall be discussed.

- **Redundancies & Survivability.** The primary, secondary, and tertiary backups and alternate modes shall be discussed. The candidate should cover concepts such as graceful degradation versus catastrophic failure for the system, battle over-ride, lockouts, emergency trips, safety interlocks, and run time on backup systems.

- **Conclusion.** The candidate should review the major system parameters with a brief reiteration of the systems overall purpose and function and provide a formal completion.

Follow-up questions shall be limited to the topics above and these additional topics.

- **Common Failure Modes.** Systems have known weak spots or failure points and quirks of operation. The candidate should be able to discuss the system beyond the academic knowledge of the CFTOs and demonstrate firsthand knowledge of common issues, fleet wide issues or historical failure trends from across the fleet.

- **Technical Administration.** The candidate should be able to speak about the system's history including topical Engineering Changes, refits, upgrades, or lifecycle management. Who the local FMF engineering subject matter experts might be. Relevant portions of SEMS, NEM, NaMMS, and other guiding documents shall be discussed if applicable. The candidate is not expected to have more than a very basic awareness of MBS, Risk Assessment and impact on operational capabilities and failures of this system as they relate to an OPDEF or capabilities. Technical Investigations, Technical Bulletins, operating restrictions, and power reductions could also be discussed here.

- **Shore support agencies, Design Authority and Operating Authority, Specialty Tools, And Test Equipment.** The candidate should be able to discuss Field Service Representative support if applicable and mention any

specialty tools or procedures for equipment such as the removal of soft patches, crane access, unique Lock Out Tag Out considerations and safety requirements or other considerations, such as docking a ship for access and repair.

Question 2 | Engineering Drills

15 minutes for question and follow-on

The board members will provide the candidate with a situation and specific drive mode. The situation is read out twice, then the emergency prompt is read.

Immediately following the prompt, the candidate shall immediately respond and identify an Engineering Problem or Engineering Emergency, and identify the specific action and considerations taken by the EWK and SEWK until the plant is safe and there is no risk of damage to the propulsion plant. For engineering problems, following the initial response from the candidate, the board can drive the scenario by denying a speed change from the OOW, or having the speed reduction fail to resolve the issue. It is expected that the candidate will answer follow-up questions on such as: What is the possible cause(s) of the failure, what might follow-on investigation involve, what drive modes are available, how long will the repairs take and what capabilities and redundancies have been lost.

Question 3 | Minor System or Damage Control System

10 minutes preparation, 10 minutes delivery, 5 minutes for follow up questions.

The board will be given the choice between a Damage Control System or other Secondary System. Similar to question 1, the candidate shall create their own drawing from memory that includes major components, layout, capabilities, limitations, operating parameters, flash up and preservation considerations, safety considerations, and emergency operation.

- **Purpose**. What is the design intent of the system and how does the system contribute to float, move, and fight.

- **Overview.** Provide a general overview of the system and orientation of the drawing. What spaces is it in? Is there a separate port and starboard, primary, and secondary system?

- **Major Components.** List and discuss the operation, location, function, and characteristics of all components,

- **Functional Description.** Describe the normal operation of the system. The suggested way to do this is to trace out the flow and pressure path in the case of fluids, or electrical signals, current flow and feedback, for data and electrical power distribution networks. Primary and alternate operation modes shall be discussed as applicable.

- **Theory, Tests and Trials.** Briefly discuss relevant engineering theories and principles used by the system. Describe and explain common and important maintenance routes, their purpose and whether completed by ship's staff, field service rep, FMF or others. This may not apply to all systems.

- **Interdependencies.** All systems rely on other systems for input and output and interoperability. The relationship, reliance, and resiliency of the topic system to other systems shall be discussed.

- **Conclusion.** The candidate should review the system parameters with a brief reiteration of the system's overall purpose and function and provide a formal close.

Question 4 | Stability

30 minutes total; 10 minutes preparation, 10 minutes delivery, 10 minutes for follow up questions.

The candidate should use time to prepare and present a diagram, chart or illustration that helps convey the fundamental theory and operational application of the topic. The Manual of Trim and Stability, as well as equations, stability criteria and stability curves shall be incorporated into the answer as applicable. The focus should be on answering the specific question posed, not merely as an opportunity to describe and explain everything the candidate knows about stability. Possible areas of discussion include but are not limited to:

- G-Z curve, what it is showing and how it is created.
- Impact of shifting weight on the G-Z Curve and ship's stability
- Icing Condition
- Free Surface vs Free Communication
- Docking
- Grounding
- Ship's Loading conditions and weight

Question 5 | Divisional Administration

20 minutes total; 5 minutes preparation, 10 minutes delivery, 5 minutes for follow up questions.

The divisional administration question examines a candidate's ability to act in a wide array of staff, administrative, technical and leadership roles across the entire CAF. The candidate shall demonstrate a working and operationally functional level of knowledge in military administration.

Question five requires a working knowledge of the DAODs, common QR&Os, NCM and Officer Career Management, disciplinary issues, and PaCE. More specific to the RCN and NTO Occupation, the candidate shall be able to discuss Naval specific doctrine, policy, and orders and what is typically covered within CANFORGENs, NAVORDS, SSOs, CSOs, COTMs, EOTMs and ship specific SOPs.

The candidate is not expected to know specific details of specific policies, but rather an emphasis is placed on a broad awareness of what the relevant references are, where to find, and how to properly execute policy to the benefit of members. Ideally a candidate can share their own experience in the topic stemming from professional experience as link in the divisional system chain within the department onboard a ship.

MS Eng Basic Occupation Qualification (BOQ) Board Marking Sheet	
Candidate:	**Ship:**
Qualification: AIRY	**Date:**
Chairperson:	**Examining Officer:**
Assessing Member:	**Divisional Representative:**

Instructions:
Each question is scored out of 10 for content and presentation which are weighted to create a score out of 10 for each question.[1] Each question is then weighted to create an overall board mark.[2] The candidate shall achieve a minimum overall board mark of 60%, including a minimum of 60% for questions 1 and 2 on both content and presentation (critical questions).

Score	**Content**	Score	**Presentation**[3]
0	Unable to answer any part of the question or even begin to discuss the topic.	0	Many major deficiencies
1	Unable to communicate a basic level of technical knowledge of engineering principles.	1	Some major deficiencies
2	Able to answer minor parts of the main question but unable to answer follow up questions.	2	Many intermediate deficiencies
3	Displays a very low level of technical knowledge of engineering principles.	3	Some intermediate deficiencies
4	Able to answer parts of the main question and a limited number of follow up questions displaying only a restricted level of technical knowledge.	4	Many minor deficiencies
5	Requires assistance answering the main question and requires prodding to answer follow up questions.	5	Some minor deficiencies
6	Displays an acceptable level of technical knowledge of engineering principles and administration requirements.	6	Acceptable
7	Displays an above average level of technical knowledge of engineering principles.	7	Good
8	Displays a high level of technical knowledge of engineering principles.	8	Very Good
9	Generates discussion through the display of excellent technical knowledge of engineering principles.	9	Excellent
10	Displays an exceptional level of technical knowledge answering the main question and follow up questions well beyond the expected level.	10	Exceptional

Note 1	Qualification	Content	Presentation
	NEI	0.8	0.2
	BOQ	0.8	0.2
	HOD	0.6	0.4

Note 2	Question	Weight
	1	5
	2	5
	3	4
	4	3
	5	3

Note 3	Presentation is an aggregate evaluation of the following: - Dress and Deportment - Professionalism - Use of Proper Terminology - Interaction with Board Members - Eye Contact - Pace and Duration - Volume and Clarity of Voice - Body Language and Movement - Confidence and Conviction in Answers - Organization of the Presentation and Answers to Questions - Clarity of Board Drawings - Use of the Board/Drawing to Assist in Explanations.

MS Eng Basic Occupation Qualification (BOQ) Board Marking Sheet					
CANDIDATE:			**SHIP:**		
QUALIFICATION: AIRY			**DATE:**		
QUESTION	**CONTENT (80%)**	**PRESENTATION (20%)**	**QUESTION TOTAL**	**WEIGHT FACTOR**	**WEIGHTED SCORE**
1				**5**	
2				**5**	
3				**4**	
4				**3**	
5				**3**	

MARK (%): $\frac{\sum Weighted\,Score}{2} =$

RESULT: **PASS / FAIL** (circle one)

COMMENTS:

NOTE: Name and signature are mandatory

Name of Board Chair

Signature

ANNEX J – ON JOB PERFORMANCE RECORD (OJPR)

(Certificate of Qualification)

Version	Revision Date	Remarks
OJPR Version 2.0	1 July 2023	Formatted OJPR

Name and Initials	Rank	Service Number

Date Training Started	
Date of Completion	

THE OJPR MUST BE COMPLETED IN ACCORDANCE WITH THE INSTRUCTIONS BELOW. INCORRECT OR INCOMPLETE OJPR WILL BE RETURNED FOR RE-SUBMISSION.

INSTRUCTIONS

1. One copy of this OJPR, hereinafter referred to as the master copy, shall be made up for each individual beginning OJT. It shall be kept and maintained by the supervisor. The Training Establishment (TE) shall make a duplicate copy for retention by the trainee for the purpose of recording and reviewing individual progress. The TE for the OJPR must maintain Training Tracker fields as outlined in Monitor Mass.

2. As the trainee completes each Performance or Enabling Check to the required standard, IAW the associated checklist in ANNEX I, the Training PO or Officer will date and sign the applicable item on the master and duplicate copies of the OJPR sheet and inform the individual of the action taken.

3. It may not be possible to conduct OJT for some POs if, for example, the necessary equipment is not fitted on a ship, is not on issue to the unit, or is otherwise unavailable. If the training PO or Officer encounters any of these issues they are directed to contact Standards at the appropriate TE for guidance.

4. If an individual demonstrates the ability to perform additional duties not specified in this QSP, an appropriate entry shall be made under "Additional Duties" at the end of the OJPR.

PROGRESS REVIEW

5. Progress review shall be carried out quarterly by the Divisional Officer and Training Officer (Officers).

PO/EO	Description	Remarks	Rank/Name/ Signature	Date	OPI
PO 001 Main Propulsion					
EO 001.01	Gas Turbines	Present to MSEO			MSEO
		System Walkthrough			MSEO
		Observe & Participate			MSEO
EO 001.02	Diesels	Present to MSEO			MSEO
		System Walkthrough			MSEO
		Observe & Participate			MSEO
EO 001.06	Transmissions	Present to MSEO			MSEO
		System Walkthrough			MSEO
		Observe & Participate			MSEO
EO 001.07	Shaftline	Present to MSEO			MSEO
		System Walkthrough			MSEO
		Observe & Participate			MSEO
	CRPP	Present to MSEO			MSEO
		System Walkthrough			MSEO
		Observe & Participate			MSEO
EO 001.08	MLO	Present to MSEO			MSEO
		System Walkthrough			MSEO
		Observe & Participate			MSEO
	FOS	Present to MSEO			MSEO
		System Walkthrough			MSEO

PO/EO	Description	Remarks	Rank/Name/ Signature	Date	OPI
		Observe & Participate			MSEO
	MSWC	System Walkthrough			MSEO
		Observe & Participate			MSEO
PO 002 Auxiliaries					
EO 002.03	Refrigeration System	System Walkthrough			MSEO
		Observe & Participate			MSEO
EO 002.04	HP Air	System Walkthrough			MSEO
		Observe & Participate			MSEO
	LP Air	System Walkthrough			MSEO
		Observe & Participate			MSEO
EO 002.05	Steering	Present to MSEO			MSEO
		System Walkthrough			MSEO
		Observe & Participate			MSEO
EO 002.06	ASWC	System Walkthrough			MSEO
		Observe & Participate			MSEO
EO 002.07	Fresh Water Production	System Walkthrough			MSEO
		Observe & Participate			MSEO
EO 002.08	Steam	System Walkthrough			MSEO
		Observe & Participate			MSEO
EO 002.09	FOF/FOT	System Walkthrough			MSEO
		Observe & Participate			MSEO

PO/EO	Description	Remarks	Rank/Name/ Signature	Date	OPI
	RAS	System Walkthrough			MSEO
		Observe & Participate			MSEO
EO 002.10	JP-5 Fuel System	System Walkthrough			MSEO
		Observe & Participate			MSEO
	RAST	System Walkthrough			MSEO
		Observe & Participate			MSEO
EO 002.11	Waste Management Systems	System Walkthrough			MSEO
		Observe & Participate			MSEO
PO 003 Controls					
EO 003.02	IPMS	Present to MSEO			MSEO
		System Walkthrough			MSEO
		Observe & Participate			MSEO
PO 004 Electrical					
EO 004.02	EPG&D	Present to MSEO			MSEO
		System Walkthrough			MSEO
		Observe & Participate			MSEO
EO 004.03	Ship's Signatures	System Walkthrough			MSEO
		Observe & Participate			MSEO
PO 005 Hull					
EO 005.02	HVAC	System Walkthrough			MSEO
		Observe & Participate			MSEO
EO 005.03	Pollution Abatement System	System Walkthrough			MSEO

PO/EO	Description	Remarks	Rank/Name/ Signature	Date	OPI
		Observe & Participate			MSEO
EO 005.04	SWS	System Walkthrough			MSEO
		Observe & Participate			MSEO
EO 005.05	Fresh Water Storage and Distribution System	System Walkthrough			MSEO
		Observe & Participate			MSEO
PO 006 Engingeering Problems					
EO 006.01	Engineering Problems Procedures (Drills)	Observe & Participate			MSEO
PO 007 Stability					
EO 007.03	Stability	Observe & Participate			MSEO
PO 009 Liquid Handling					
EO 009.01	Liquids	Describe			MSEO
EO 009.02	Liquid Handling	System Walkthrough			MSEO
		Observe & Participate			MSEO
PO 010 Docking					
EO 010.01	Docking Procedure	Describe			MSEO
		Observe & Participate			MSEO
PO 011 Safety & Environment					
EO 011.01	Safety & Environmental Management Programs	Participation			MSEO
Other Requirements					
Supplemental Instructions	40 Sea Days	Participation			MSEO
		Roundsperson			MSEO
Supplemental Instructions	Engineering Watchkeeping	DCC OP			MSEO
		ETOW			MSEO

PO/EO	Description	Remarks	Rank/Name/ Signature	Date	OPI
		MCCOP/Eng Wk			MSEO
		EOOW/ Sr Wk			MSEO
		Ops			MSEO
		Bridge			MSEO
		ERT			MSEO
Supplemental Instructions	Departmental Organization	Describe			MSEO
Supplemental Instructions	Drawing Package	To include all major systems asked on final board			MSEO

Progress Review Record

Authority & Period	Reviewed By	Signature	Date
Divisional Officer (Q1)			
Training Officer (Q1)			
Divisional Officer (Q2)			
Training Officer (Q2)			
Divisional Officer (Q3)			
Training Officer (Q3)			
Divisional Officer (Q4)			
Training Officer (Q4)			

TRAINING COMPLETION CERTIFICATE

I certify that_____has completed training to the standards stated in QSP – MS Eng BOQ, except as noted, and is recommended to challenge the Basic Officer Qualification Board.

Date training completed:	
MSEO	Date
XO	Date
CO	Date

CERTIFICATION OF QUALIFICATION

I certify that _____ has completed the OJT to the standards stated in the Marine Systems Engineering Basic Officer Qualification QSP..

TE Cmdt/CO	Date

ANNEX K – TT PLAN

This annex was not present in previous QSP.

EO	**COA 1** **TEL Delivery**	**COA 2** **Interim Delivery**	**Impact if TEL not implemented** - Realized benefits - E/W Coast Delta

ANNEX L – RECORD OF DECISIONS TPWB

1. This record summarizes the conduct of the TPWB including major activities, important discussions and contentious issues, as well as all decisions and recommendations made by the Board. A description of the reasoning which supports a decision or recommendation is provided when required.

2. With the new format of the QSP, this information was not found in the previous QSP. The qualification standard record of decisions has some specific about the TPWB.

RECORD OF AMENDMENT & AMPLIFICATION

1. The record of amendment and amplification will be filled in whenever a change to the TP is initiated. This section is used to provide amplification details and provide a history of the changes made. In the requirement for the amendment section, the member inputting the changes will transpose the information from the submitted TP Change Request (TPCR), thus ensuring the QSP has the most relevant information.

Version:	**2.0**
Date: 1 July 2023	**Who Initiated Change: HMCS Venture**
Requirement for the Amendment: Rewrite of QSP in new RCN QSP Format	
Description of Changes: The QSP has been rewritten In the new RCN QSP Format. There were no changes made to any EO/PO content. The qualification board was changed to introduce drawings back into the board. Annex I was also edited for more clear direction on what is required during the board.	
Change Tracking number:	
Trg Mgr: Lt(N) Newcombe	

ANNEX M – ACRONYMS AND ABBREVIATIONS

1. This annex is to be utilized only to identify uncommon abbreviations used within this QSP.

ACRONYM / ABBREVIATION	TERM/DEFINTIONS
ALARP	As Low as Reasonably Practicable
BOI	Board of Inquiry
BOQ	Basic Officer Qualification
CANAVMOD	Canadian Naval Modification
CCA	Curriculum Control Authority
CFQAR	Canadian Forces Quality Assurance
CRCN	Commander Royal Canadian Navy
CRR	Combat Readiness Requirements
CNPTG	Commander Naval Personnel Training Group
DGMEPM	Director General Materiel Engineering Procurement Management
DGMFD	Director General Maritime Force Development
DMCM	Director Maritime Class Management
DMFM	Director Maritime Fleet Management
DMP	Directed Maintenance Period
DMS	Defence Management System
DMSS	Director Maritime Ship Support
DRDC	Defence Research Development Canada
DRMIS	Defence Resource Management Information System
DRP	Directed Repair Period
DSP	Defence Services Program
EC	Engineering Change
EHM	Equipment Health Monitoring
EMREQ	Emergency Requirement
ESL	Equipment Support Lists
EWP	Extended Work Period
FEL	Fitted Equipment List
FMF	Fleet Maintenance Facility
FSR	Fleet Support Representative
FTA	Fleet Technical Authority
HPR	High Priority Request
HRI	Harbour Readiness Inspections
ILS	Integrated Logistic Support
LCMM	Life Cycle Materiel Manager
MATREQ	Materiel Requirement
MMPT	Main Machinery Power Trial
NaMMs	Naval Materiel Management System
NAVORDs	Naval Orders
NDID	Naval Defence Identification Directory

ACRONYM / ABBREVIATION	TERM/DEFINTIONS
NEC	Naval Engineering Common
NEM	Naval Engineering Manual
NESTR	Naval Electronics Test Range
NETE	Naval Engineering Test Establishment
NFP	Notice For Power
NFS(A / P)	Naval Fleet School (Atlantic / Pacific)
NPTG	Naval Personnel Training Group
OCCAP	Oil and Coolant Condition Analysis Program
OEM	Original Equipment Manufacturer
OJPR	On-Job Performance Requirements
OPDEF	Operational Deficiencies
OPEVAL	Operational Evaluation
OPSKED	Operational Schedule
OT Div	Officer Training Division
PAYE	Payable at Year End
PEL	Periodic Engineering Letter
PIF	Pre Installation Failure
PLAR	Prior Learning Assessment and Recognition
PWGSC	Public Works Government Services Canada
QETE	Quality Engineering Test Establishment
QSP	Qualification Standard and Plan
RRI	Reduced Readiness Inspections
RSP	Readiness and Sustainment Program
SEMS	Safety and Environmental Management System
SHIPALT	Ship Alteration
SHIPMOD	Ship Modification
SI	Summary Investigation
SITREP	Situation Report
SMP	Self Maintenance Period
SOCD	Statement of Capability Deficiency
SWP	Short Work Period
TDP	Technical Data Package
TECHVAL	Technical Evaluation
TI	Technical Investigation
TRANREQ	Transfer Request
TRB	Training Review Board
TRP	Tiered Readiness Program
UCR	Unsatisfactory Condition Report
VA	Vibration Analysis

ANNEX N – DESIGNATED TRAINING ESTABLISHMENTS AND EOs SPLIT BY MITE

MITE	109394	TBD	TBD	TBD	109395	109396
TE	Sultan	HMCS Venture / NFS(A)	HMCS Venture / NFS(A)	HMCS Venture / NFS(A)	HMC Ships	HMCS Venture / NFS(A)
Phase	Applications	Ashore Phase	Stability	EODC	Afloat Phase	Qualification Board
PO	EOs					
001	001.01, 001.02, 001.03, 001.04, 001.05, 001.06, 001.07, 001.08, 001.09	001.01, 001.02, 001.03, 001.04, 001.05, 001.06, 001.07, 001.08, 001.09			001.10	
002	002.01, 002.02, 002.03, 002.04, 002.05, 002.06, 002.07, 002.08, 0020.9, 002.10, 002.11, 002.12, 002.13	002.01, 002.02, 002.03, 002.04, 002.05, 002.06, 002.07, 002.08, 002.09, 002.13			002.13	
003	003.01, 003.02, 003.03,	003.01, 003.02, 003.03			003.02	
004	004.01, 004.02, 004.03, 004.04	004.01, 004.02, 004.03, 004.04, 004.05			004.05	

005	005.01, 005.02, 005.03, 005.05, 005.07	005.01, 005.02, 005.03, 005.04, 005.05, 005.06, 005.07			005.02, 005.03, 005.04, 005.05, 005.06, 005.07	
006					006.02	
007			007.01, 007.02, 007.03, 007.04, 007.05, 007.06		007.03	
008				008.01	008.02	
009	009.01	009.01			009.02	
010		10.1			010.01	
011		011.01			011.02	
PC						PC 001